Oral Defense

Quadrature-VCO Design and 5GHz Phase-Locked Loop with Controlled Delay Line

四相位壓控振盪器設計與具延遲電路調整之 5GHz 鎖相迴路



2022.7.20

研究生 ： 王維碩

指導教授 ： 林嘉洋 博士

ADVANCED CIRCUITS AND SYSTEM ON CHIP (SOC) LABORATORY

Graduate Institute of Electrical Engineering

National Taipei University

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- Conclusion & Reference

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- Conclusion & Reference

5G New Radio in Release 16[15][16]

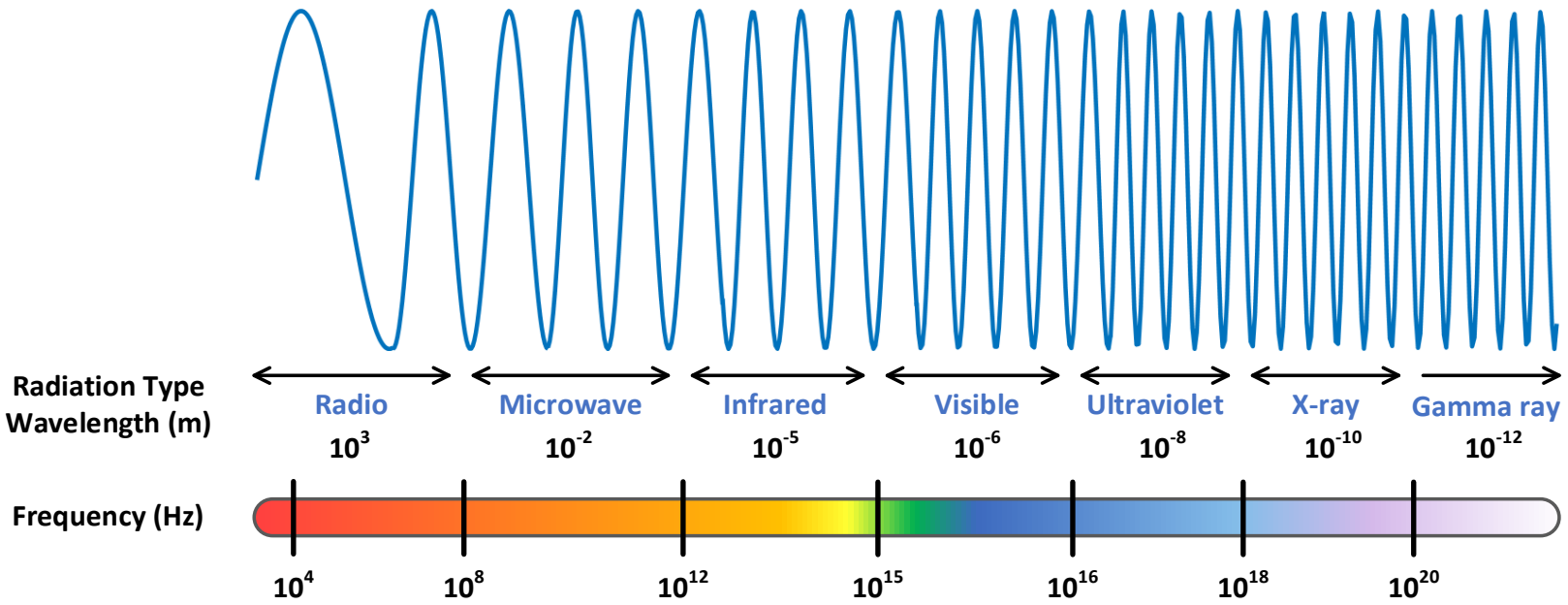


Fig 1. Electromagnetic waves

Tab 1. 3GPP 5G NR FR1 add to Release 16

Frequency band	f（MHz）	Upload（MHz）	Download（MHz）
n46	5200	5150 – 5925	
n96	6000	5925 – 7125	

Contents

- Introduction
- **Related Technology**
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- Conclusion & Reference

Superheterodyne Receiver Structure

Local Oscillator

- Generate a stable signal for mixers for mixing frequency.
- If the signal is not stable, it will be easily changed by noise and the frequency shake is too large.
- Providing a stable LO signal requires the use of a phase-locked loop system architecture.

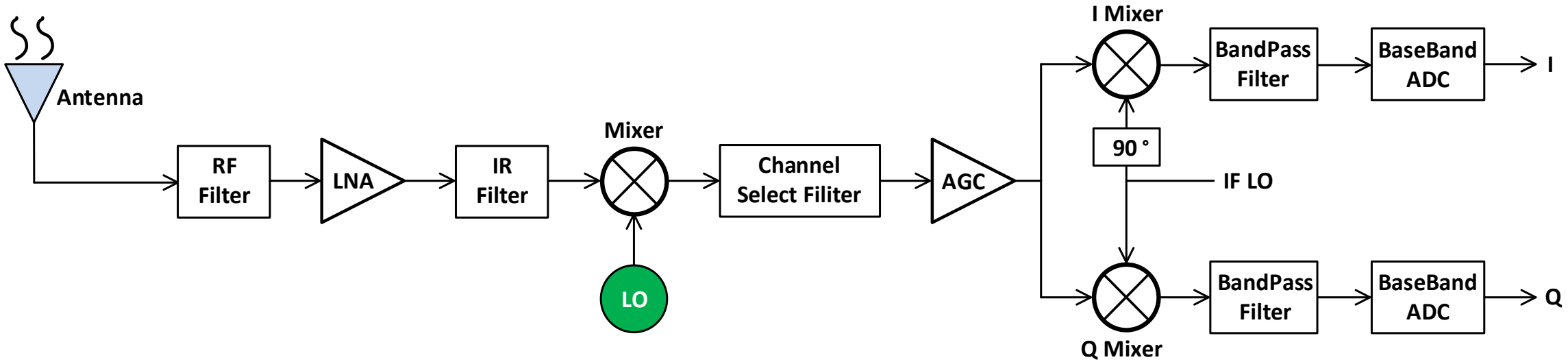


Fig 2. Structure of superheterodyne receiver

Phase-Locked Loop

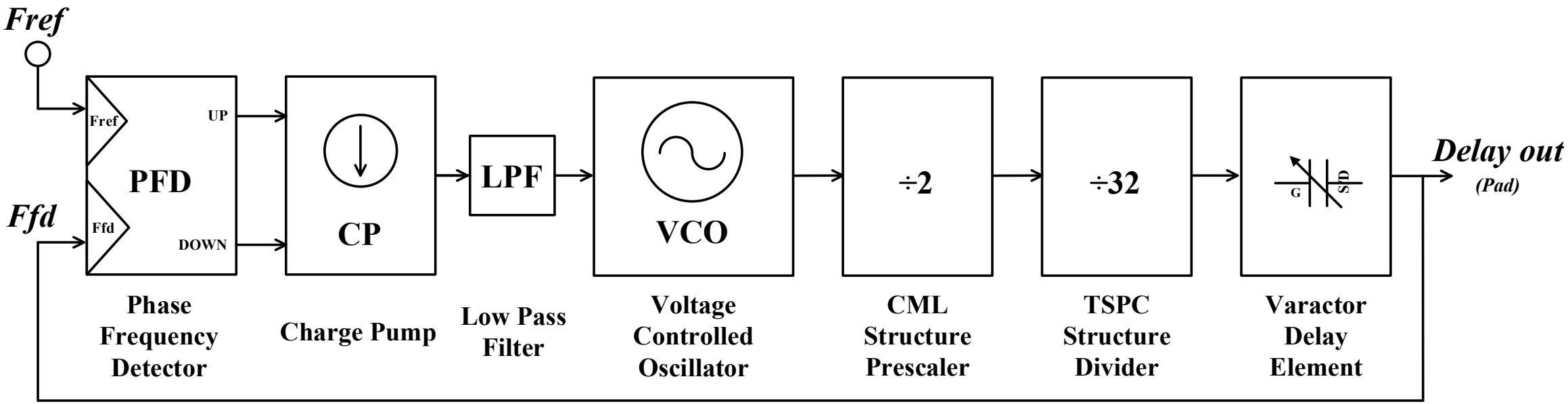


Fig 3. System of phase-locked loop

Contents

- Introduction
- Related Technology
- **Phase Frequency Detector & Charge pump**
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- Conclusion & Reference

Phase Frequency Detector

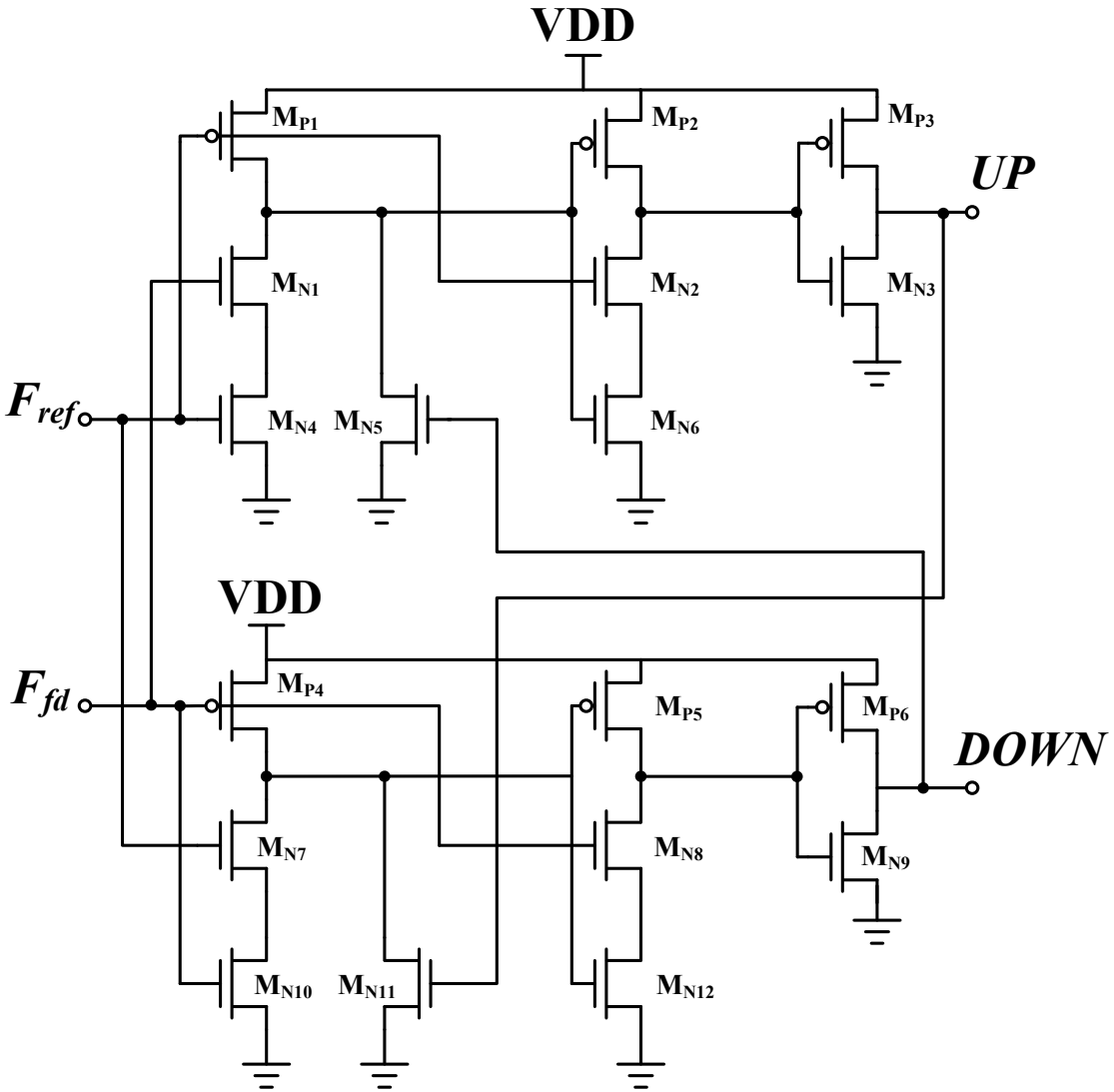


Fig 4. Structure of phase frequency detector

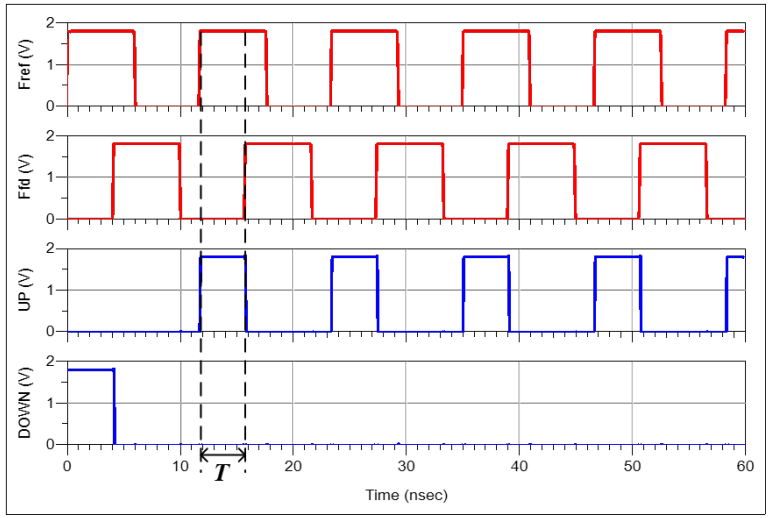


Fig 5. F_{ref} lead to F_{fd}

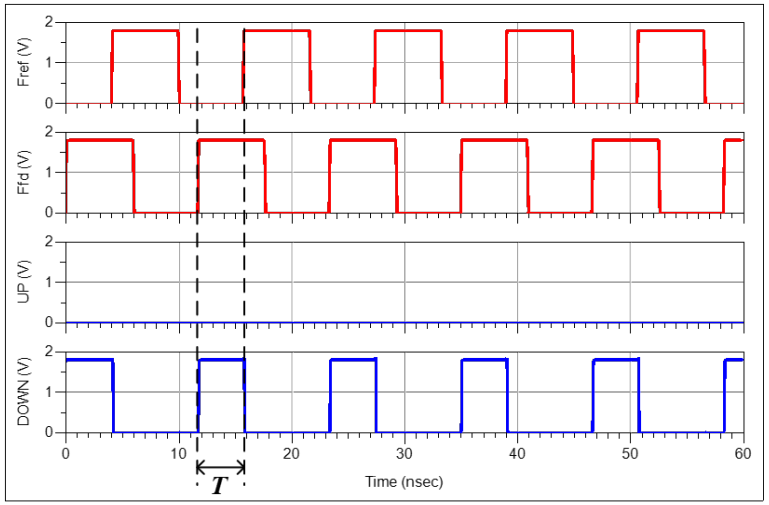


Fig 6. F_{fd} lead to F_{ref}

Phase Frequency Detector

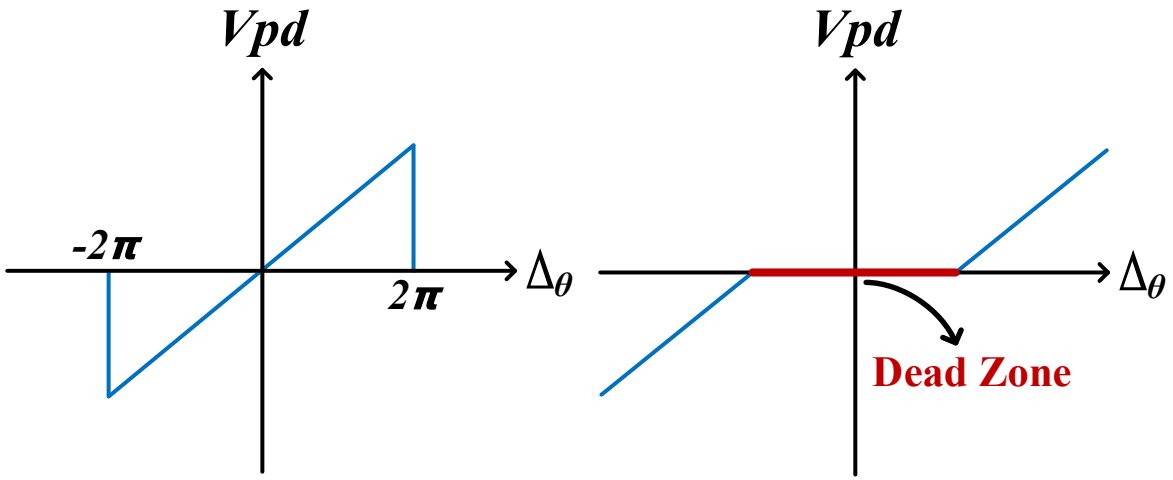


Fig 7. Dead Zone

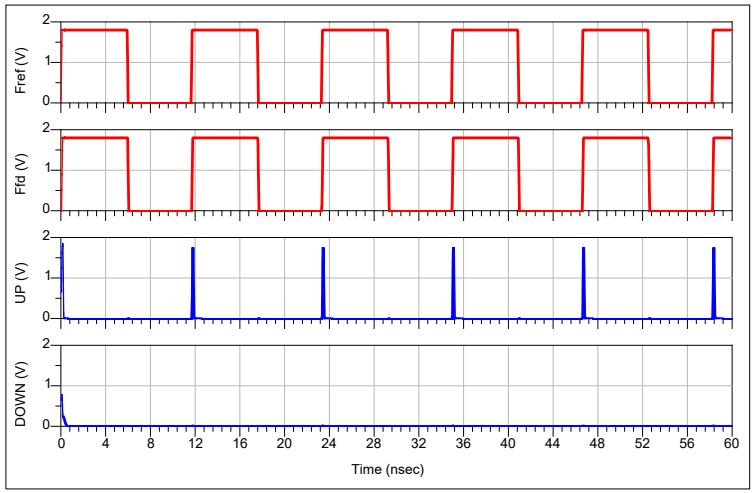


Fig 8. F_{ref} lead 0.05 nsec to F_{fd}

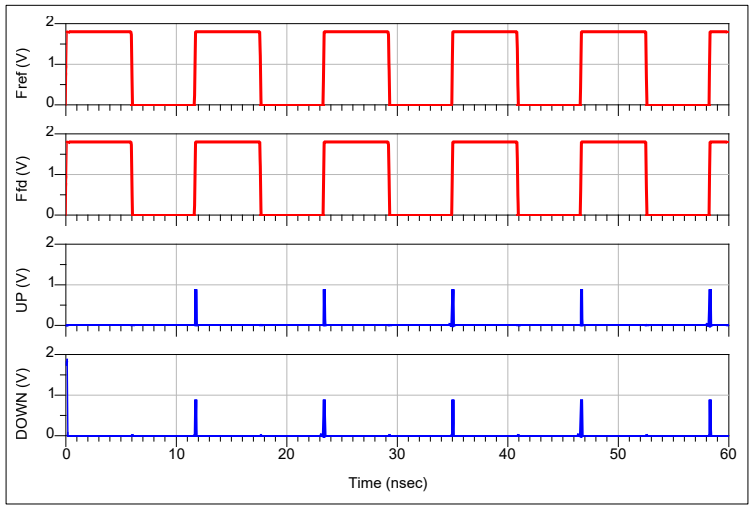


Fig 9. F_{ref} equal to F_{fd}

Charge Pump

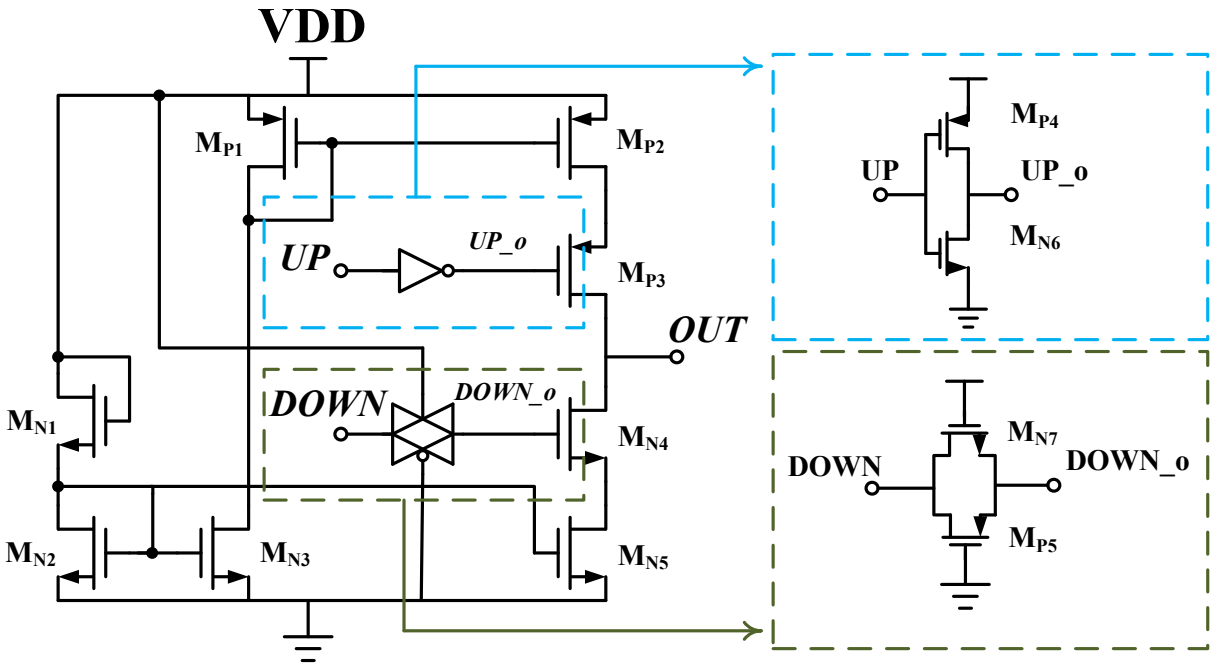


Fig 10. Structure of Charge Pump

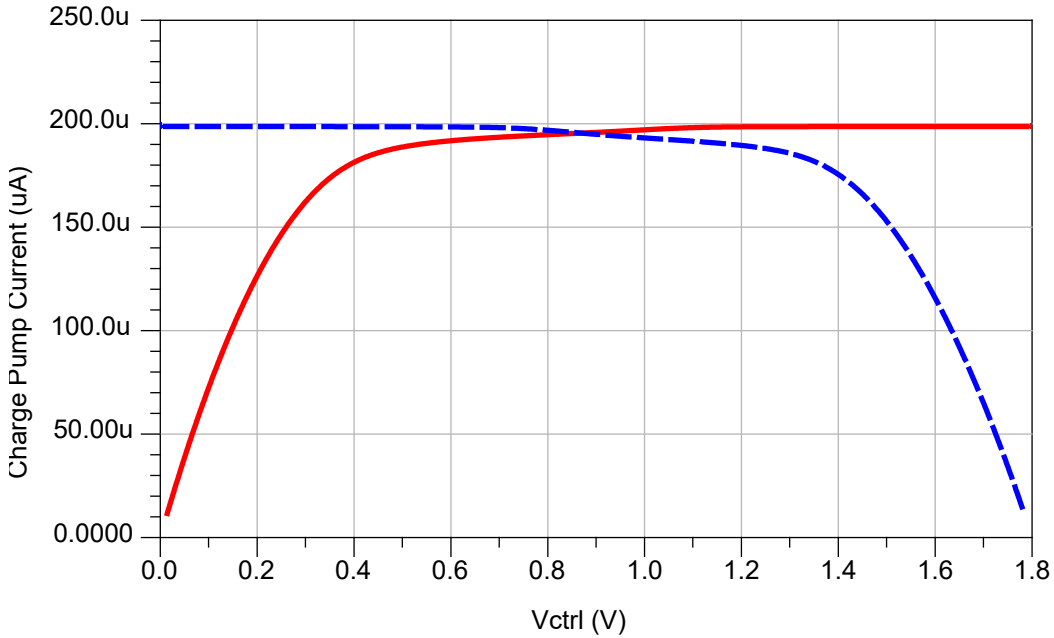


Fig 11. Charge pump charge/discharge current curve

Charge pump

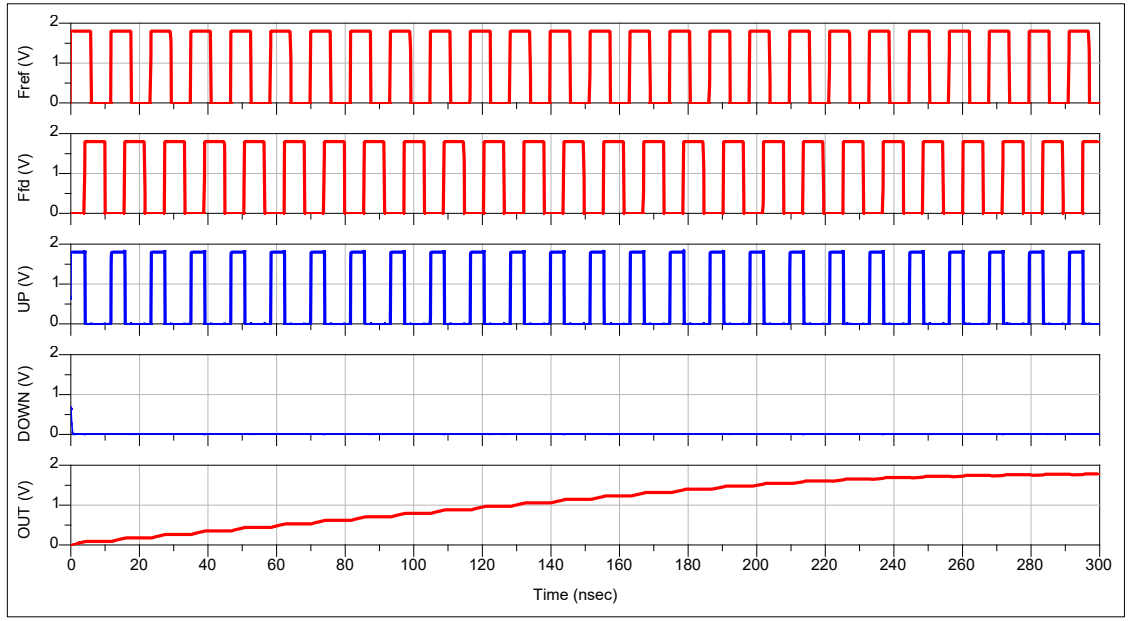


Fig 12. Charge

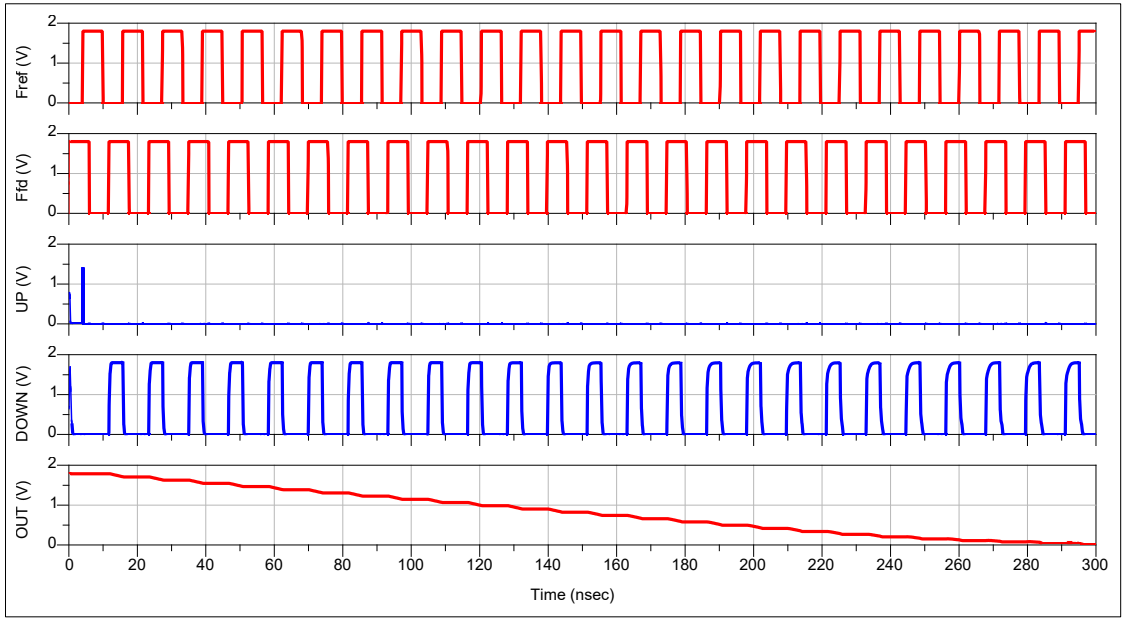


Fig 13. Discharge

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- **Loop Filter**
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- Conclusion & Reference

Structure of low-pass filter

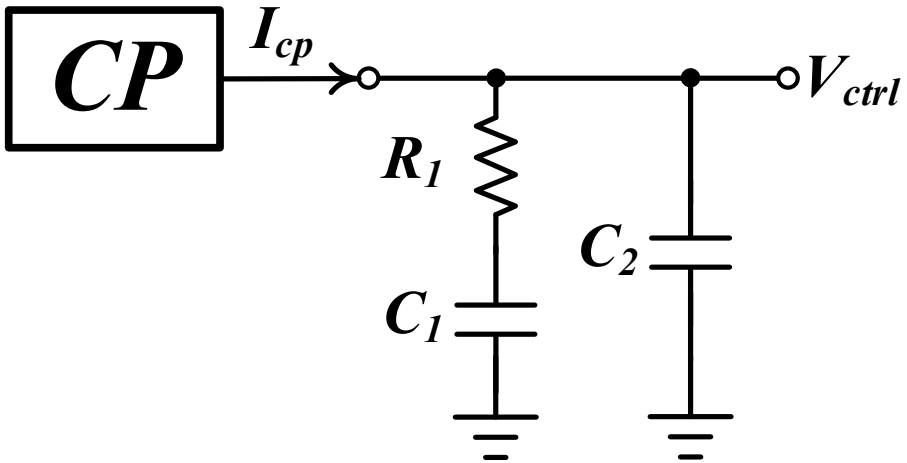


Fig 14. Structure of low-pass filter

$$Z(s) = K_{lpf} \cdot \frac{s + \omega_2}{s \cdot \left(1 + \frac{s}{\omega_3}\right)}$$

$$K_{lpf} = \frac{R_1 C_1}{C_1 + C_2}$$

$$\omega_3 = \frac{C_1 + C_2}{R_1 C_1 C_2} = \omega_2 \cdot \left(1 + \frac{C_1}{C_2}\right)$$

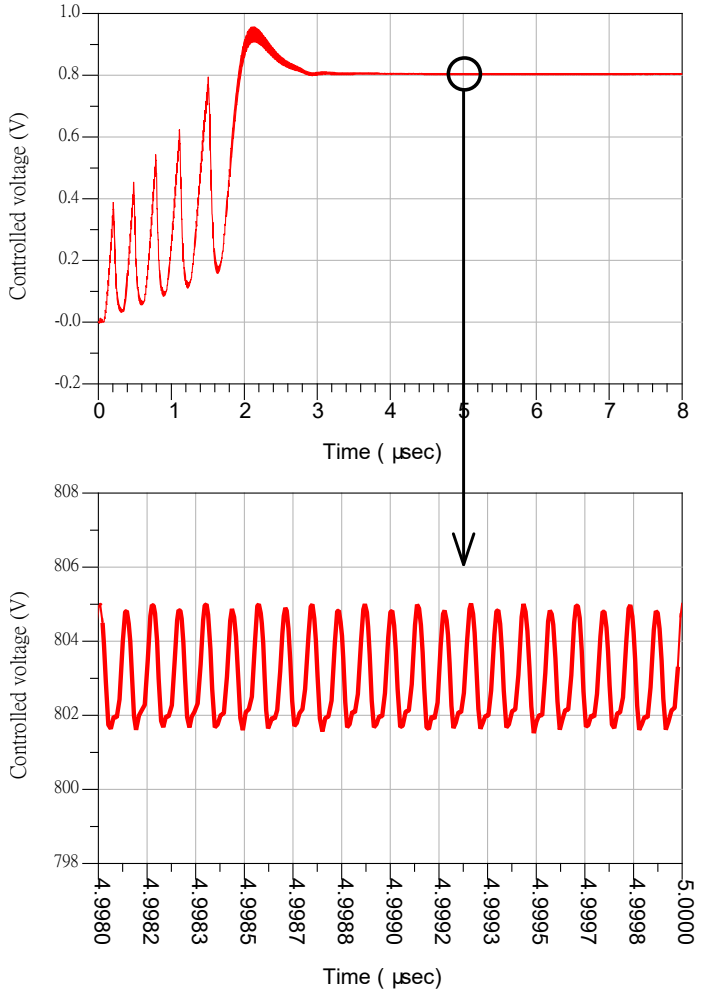


Fig 15. Function of low-pass filter

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- **Voltage Controlled Oscillator**
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- Conclusion & Reference

Structure of Voltage Controlled Oscillator

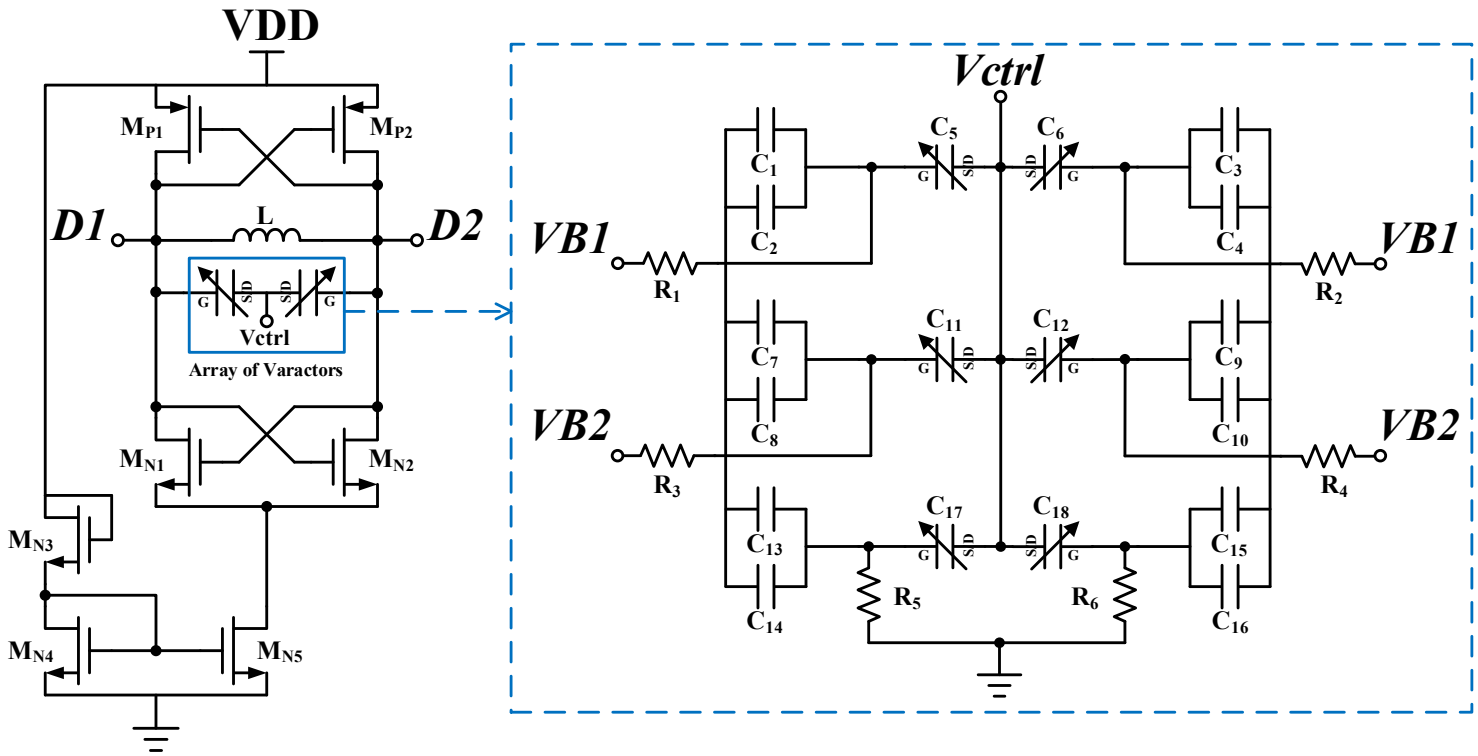


Fig 16. Structure of voltage-controlled oscillator

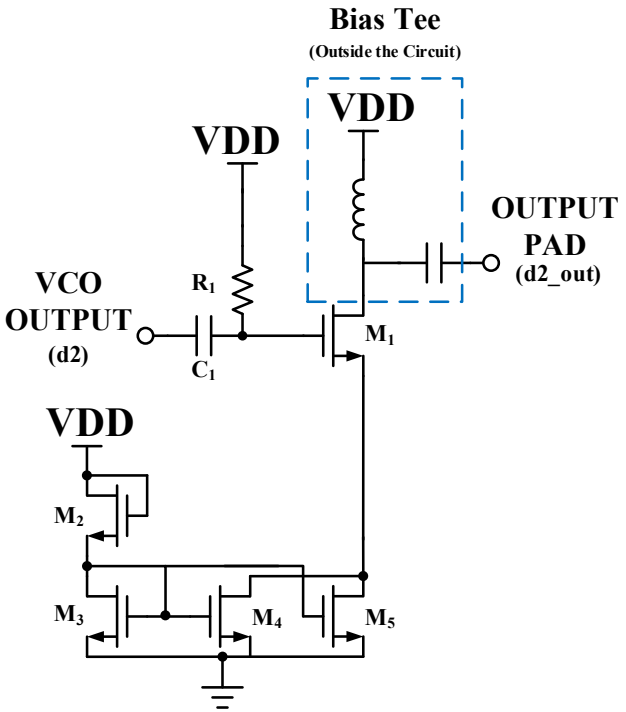


Fig 17. VCO buffer circuit

Voltage Controlled Oscillator Output Signal

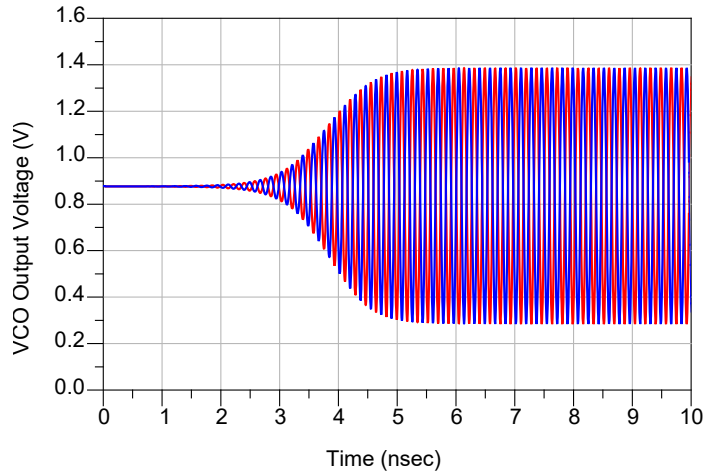


Fig 18. VCO circuit start-up time

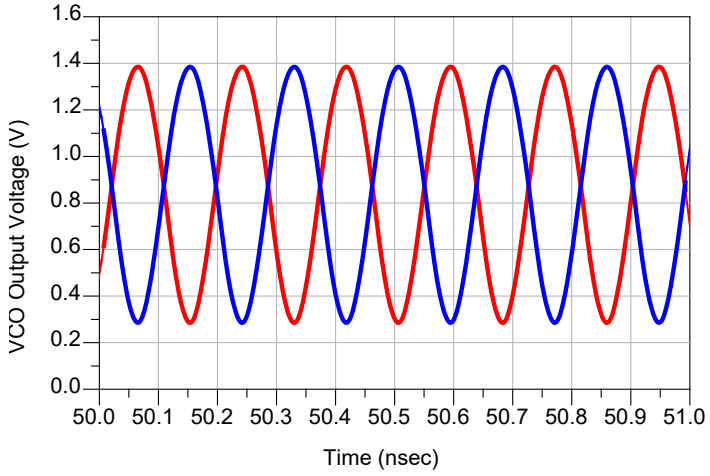


Fig 19. VCO circuit output

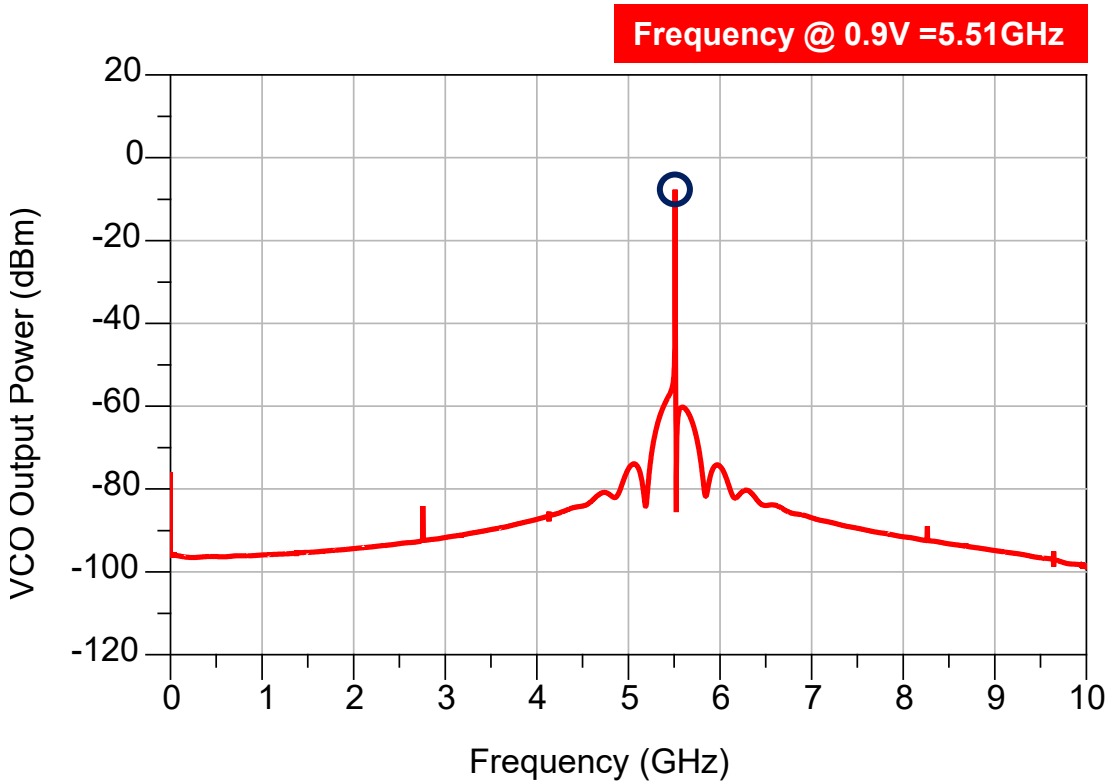


Fig 20. VCO circuit output after Fourier transform

Voltage Controlled Oscillator Output Signal

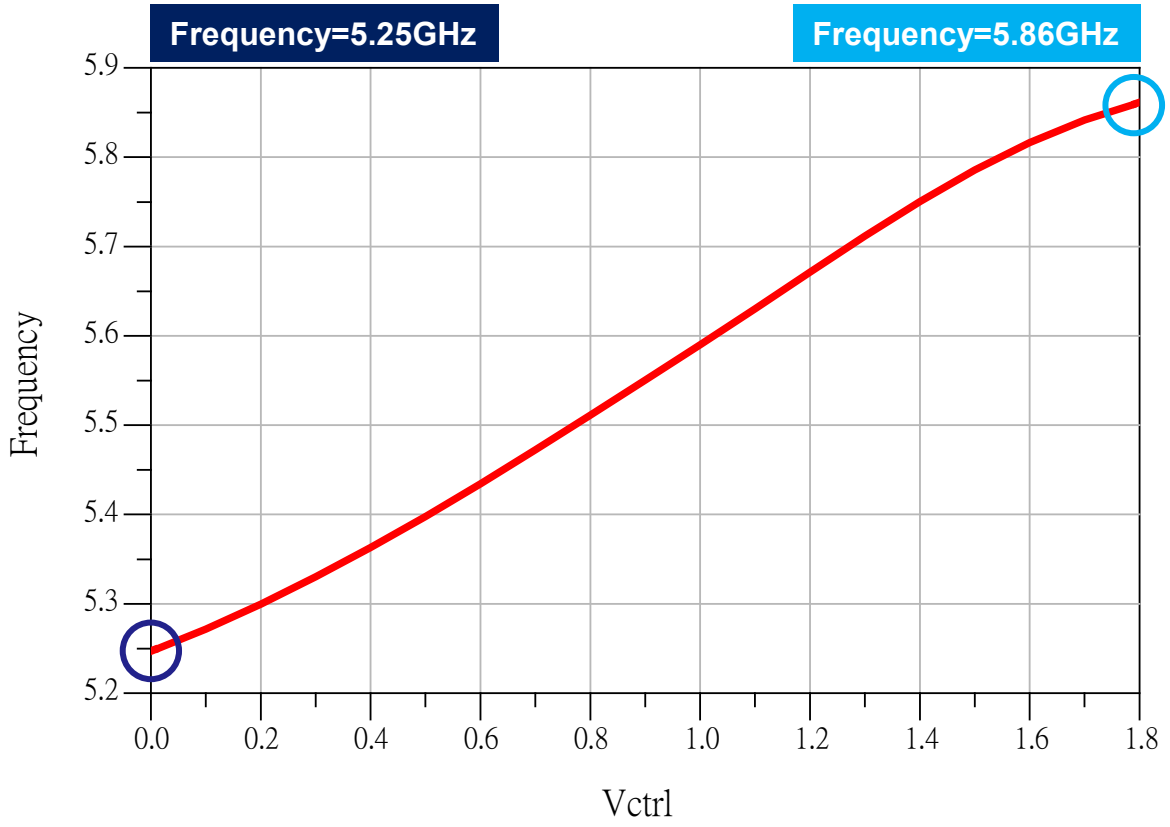


Fig 21. Tuning range of VCO circuit

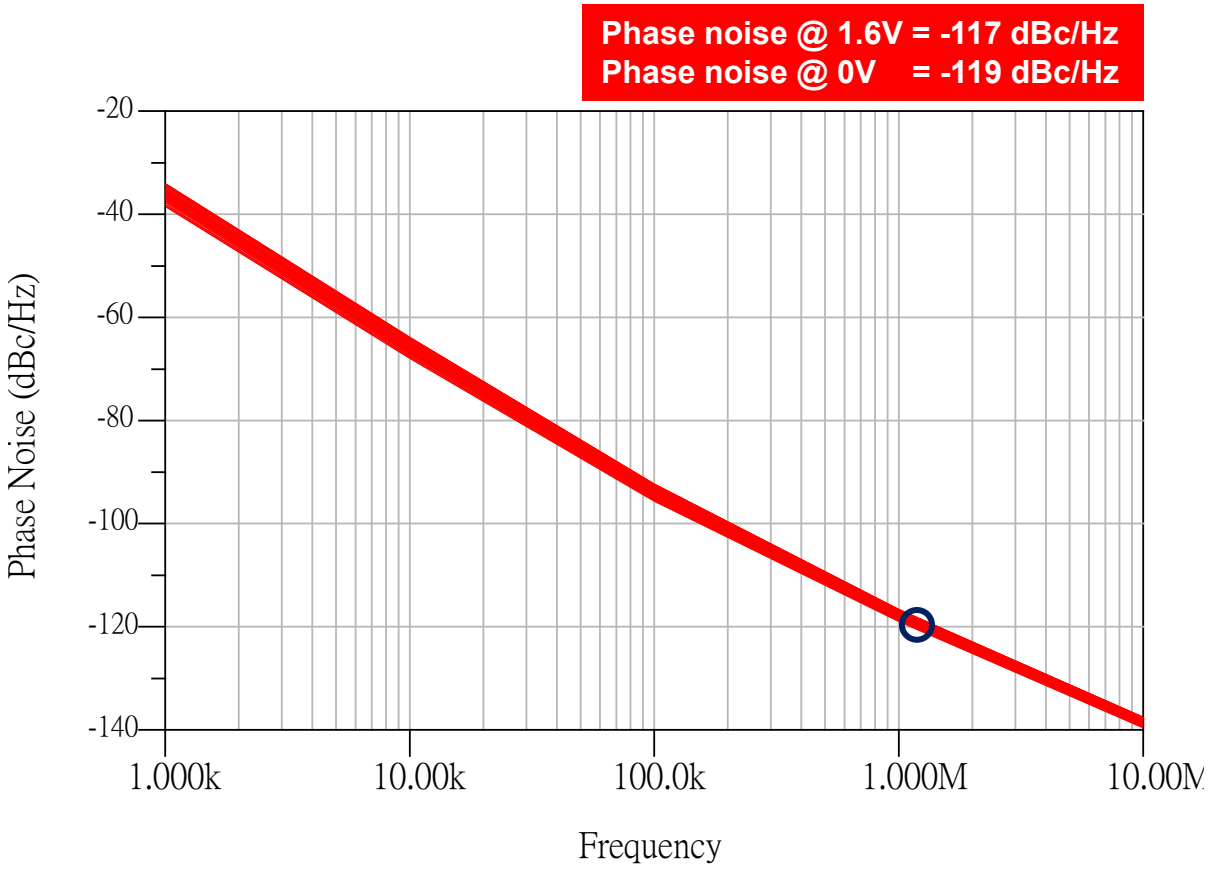


Fig 22. Phase noise of VCO circuit

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- **Quadrature-Voltage Controlled Oscillator**
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- Conclusion & Reference

Structure of Quadrature Voltage Controlled Oscillator

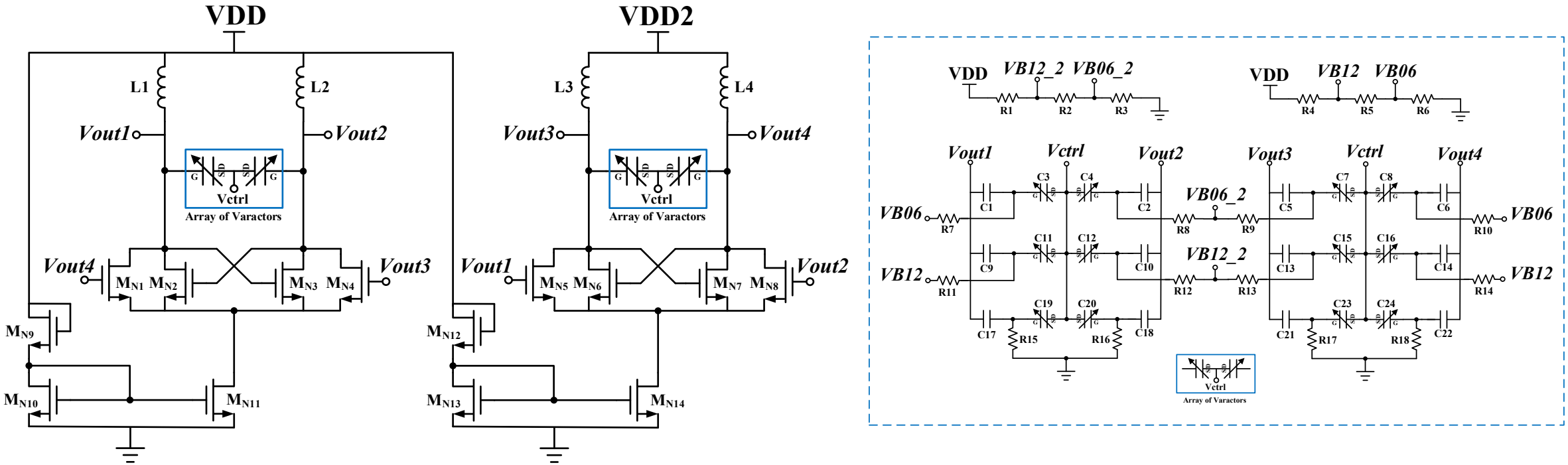


Fig 23. Structure of quadrature voltage controlled oscillator

Simulation of QVCO

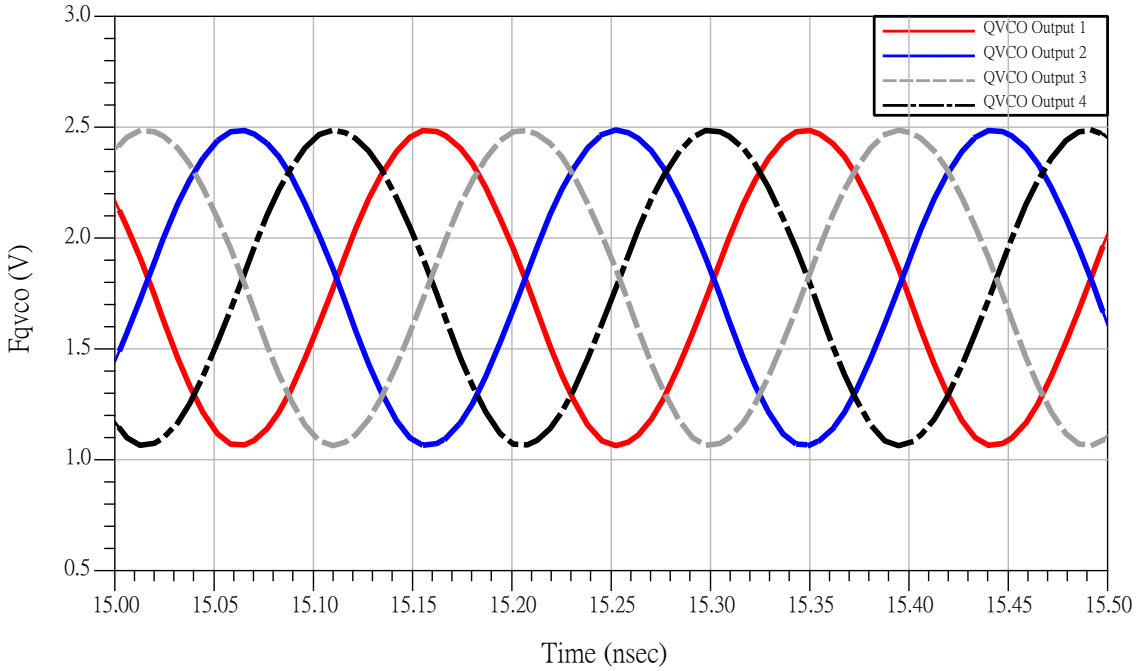


Fig 24. QVCO circuit output

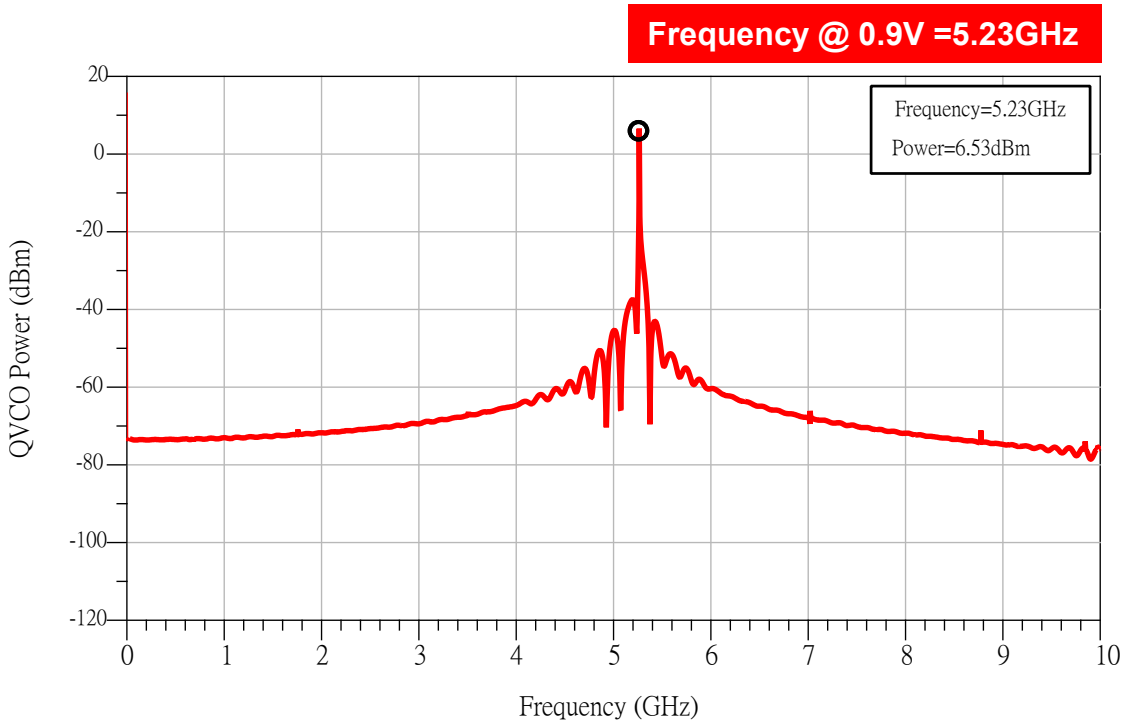


Fig 25. QVCO circuit output after Fourier transform

Simulation of QVCO

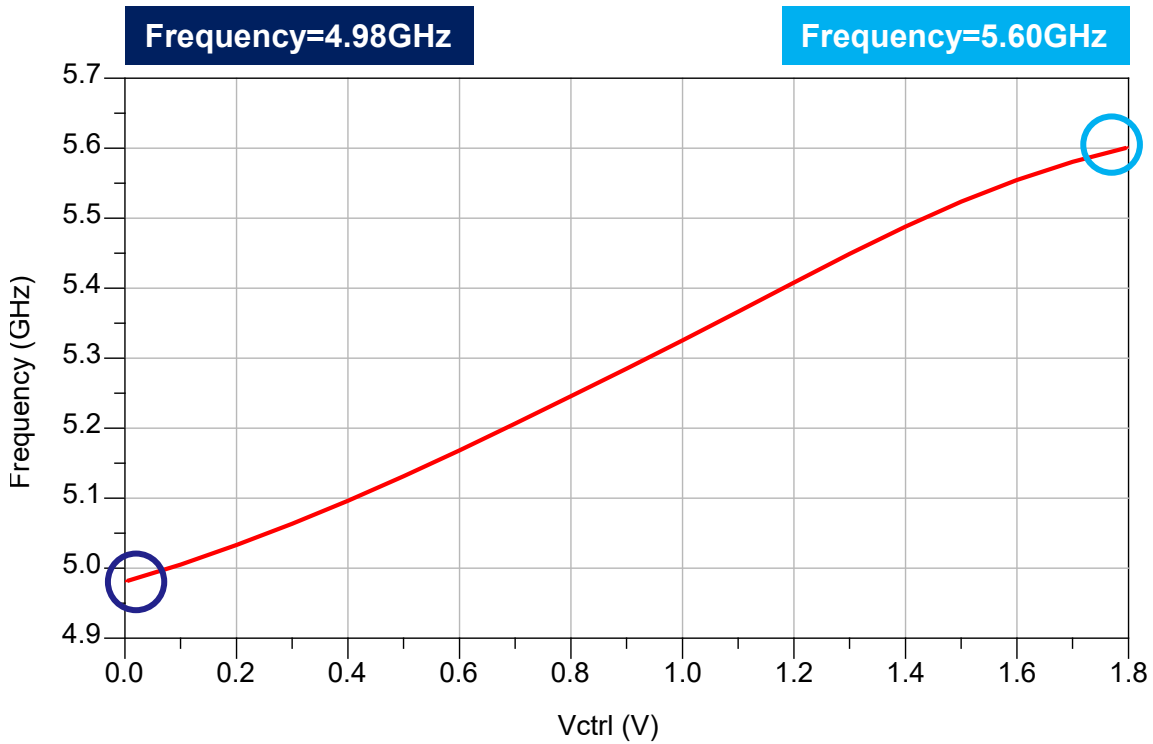


Fig 26. Tuning range of VCO circuit

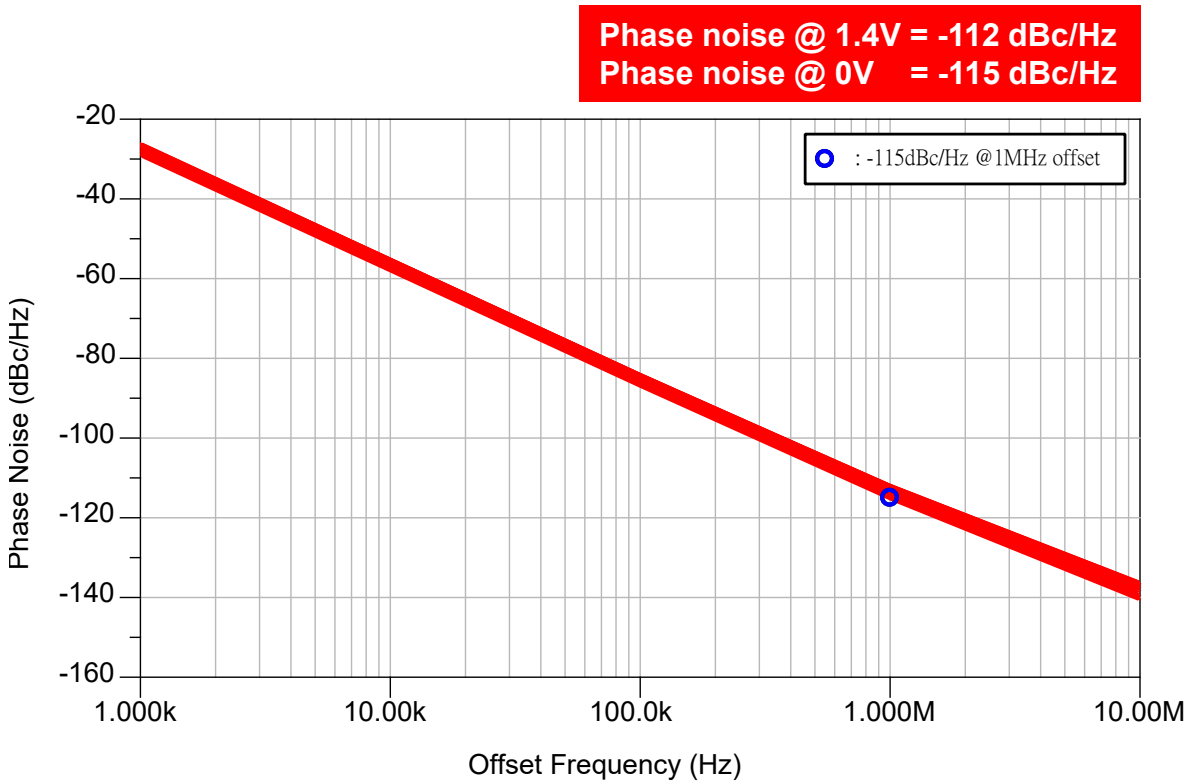


Fig 27. Phase noise of QVCO circuit

Phase Noise

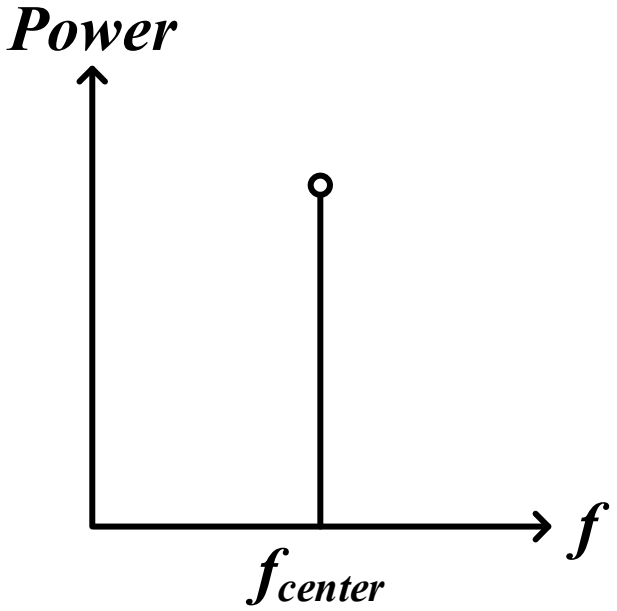


Fig 28. VCO circuit idea output signal

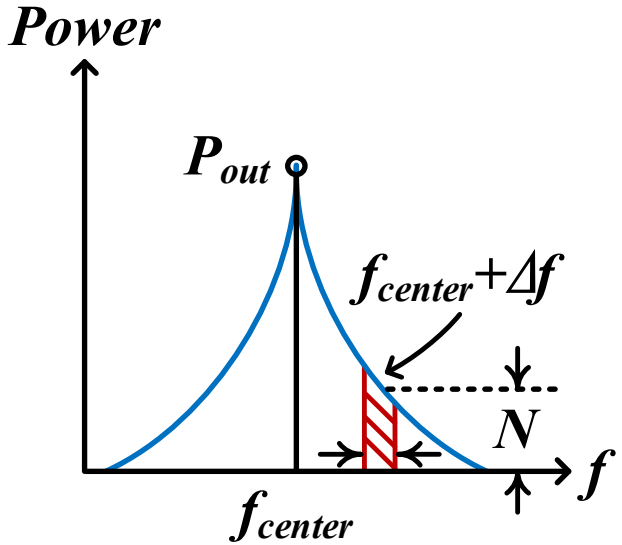


Fig 29. Real output signal

$$L\{\Delta f\} = \frac{f_{center} + \Delta f \text{ power at } 1MHz \text{ bandwidth}}{\text{carrier power}}$$

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- **Frequency Divider**
- Varactor Delay Element
- Layout & Measurement
- Conclusion & Reference

Structure of Prescaler

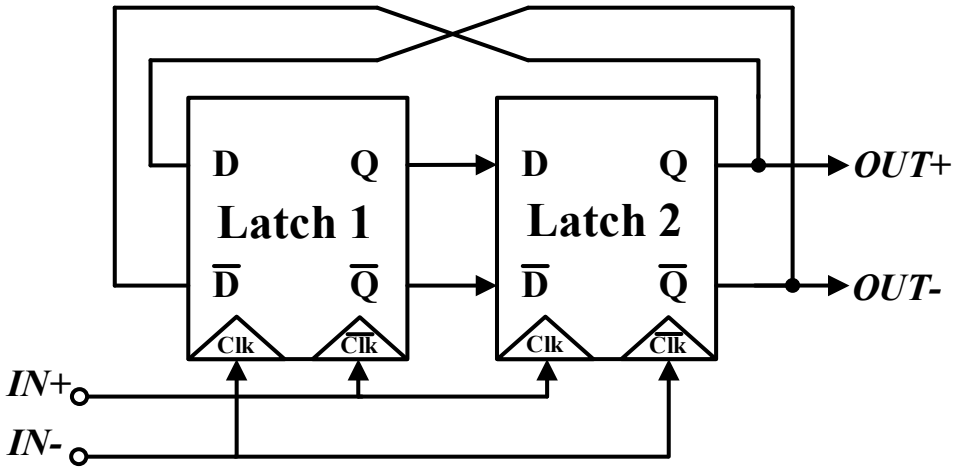


Fig 30. Structure of prescaler

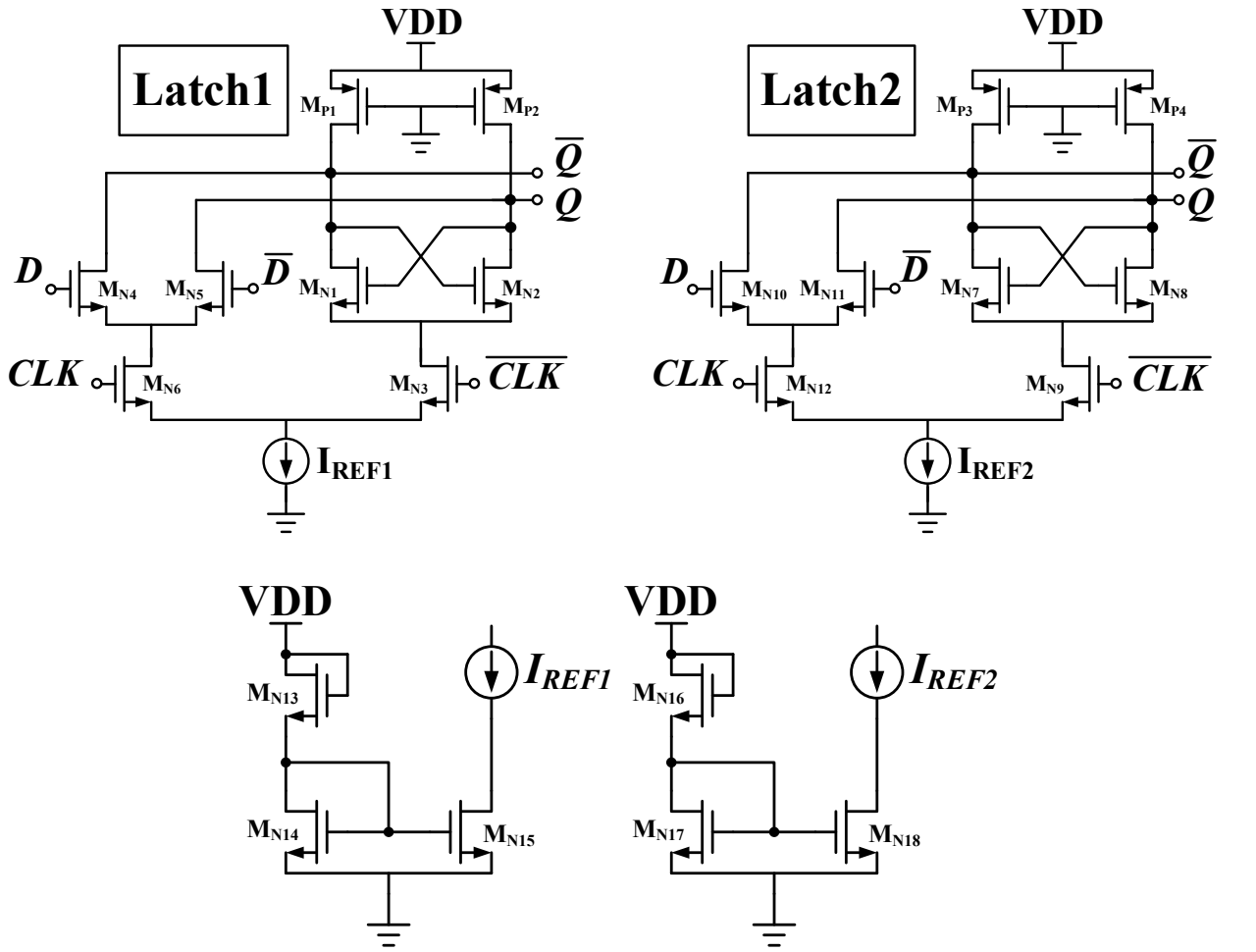


Fig 31. Detail structure of prescaler

Prescaler Simulation

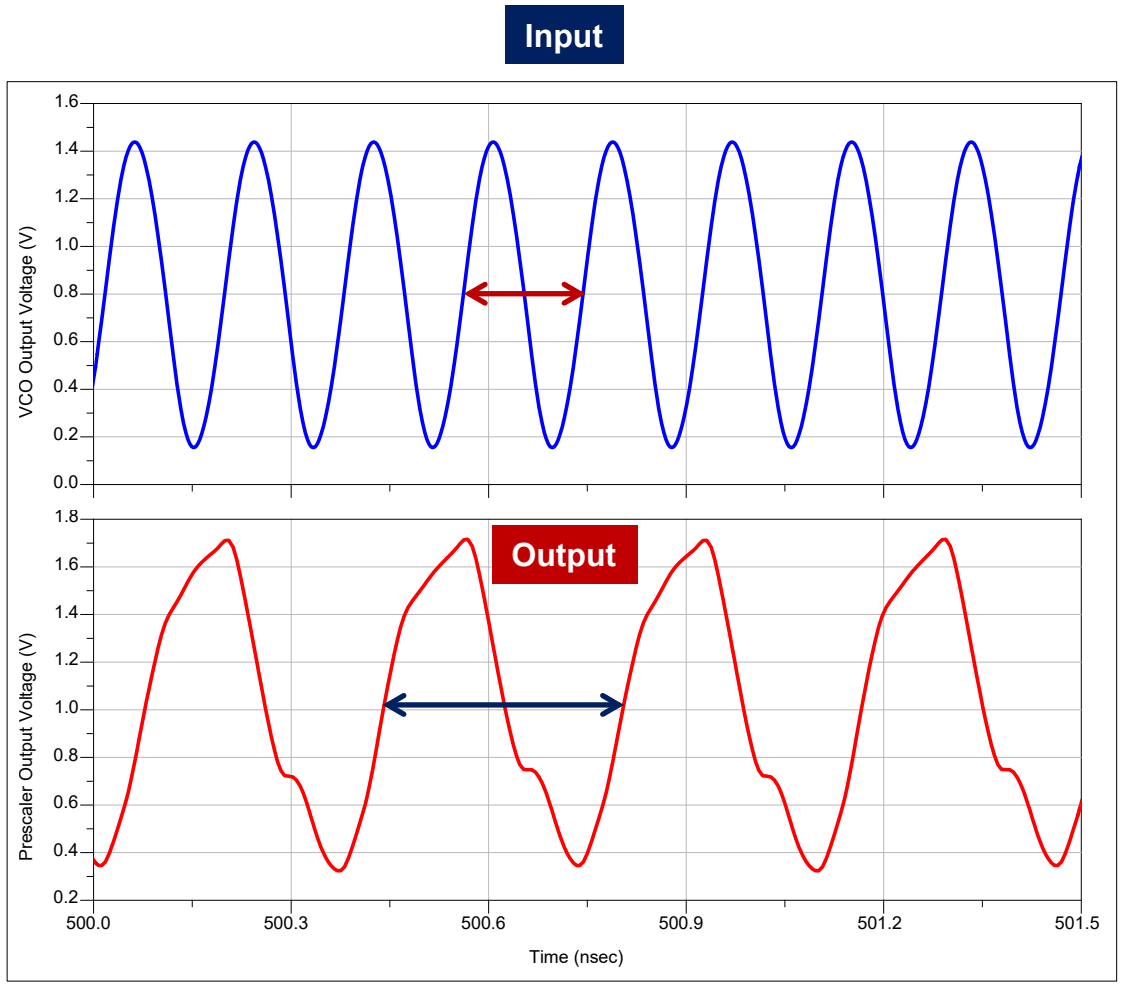


Fig 32. Input and output of prescaler

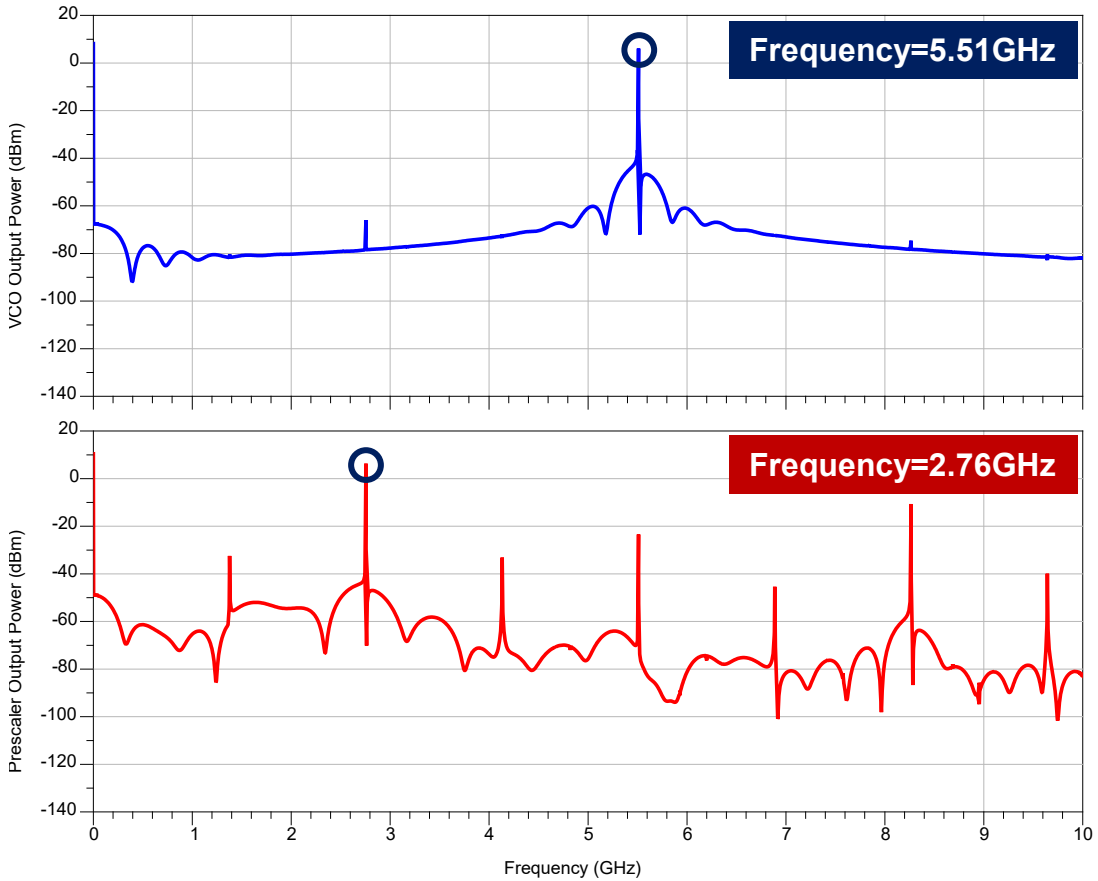


Fig 33. Input and output of prescaler with Fourier transform

Structure of True Single Phase Clock Divider

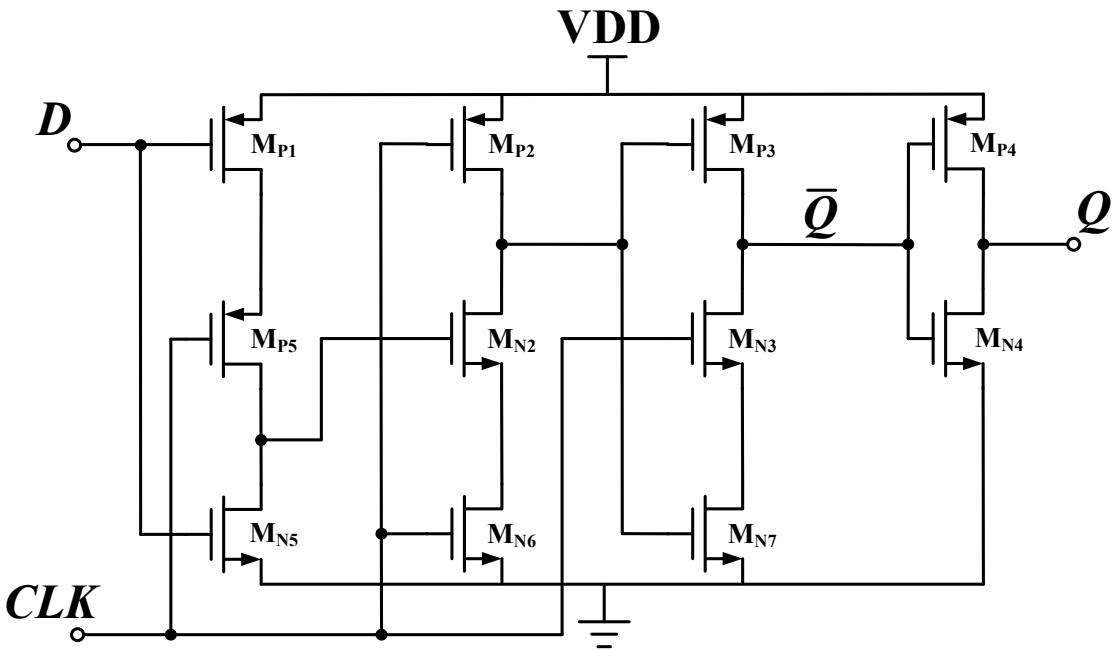


Fig 34. TSPC divider circuit

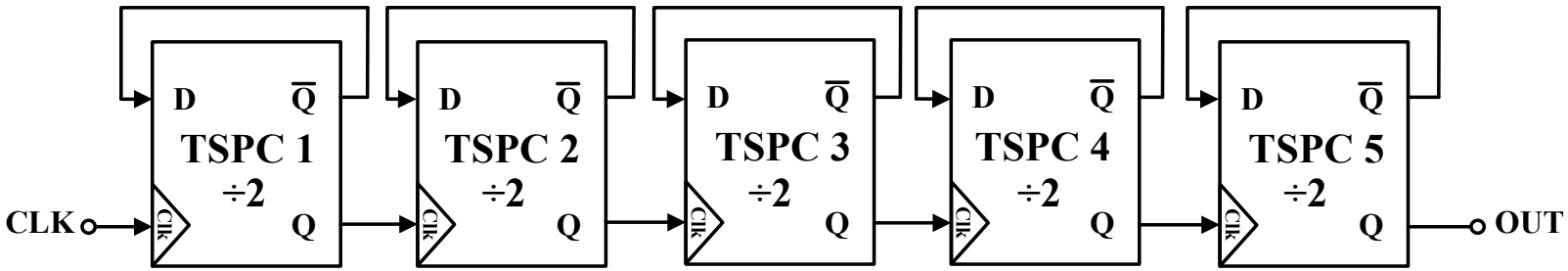


Fig 35. Chain structure of TSPC divider

TSPC Divider Chain Output

÷2
1.38 GHz

÷4
689 MHz

÷8
344 MHz

÷16
172 MHz

÷32
86 MHz

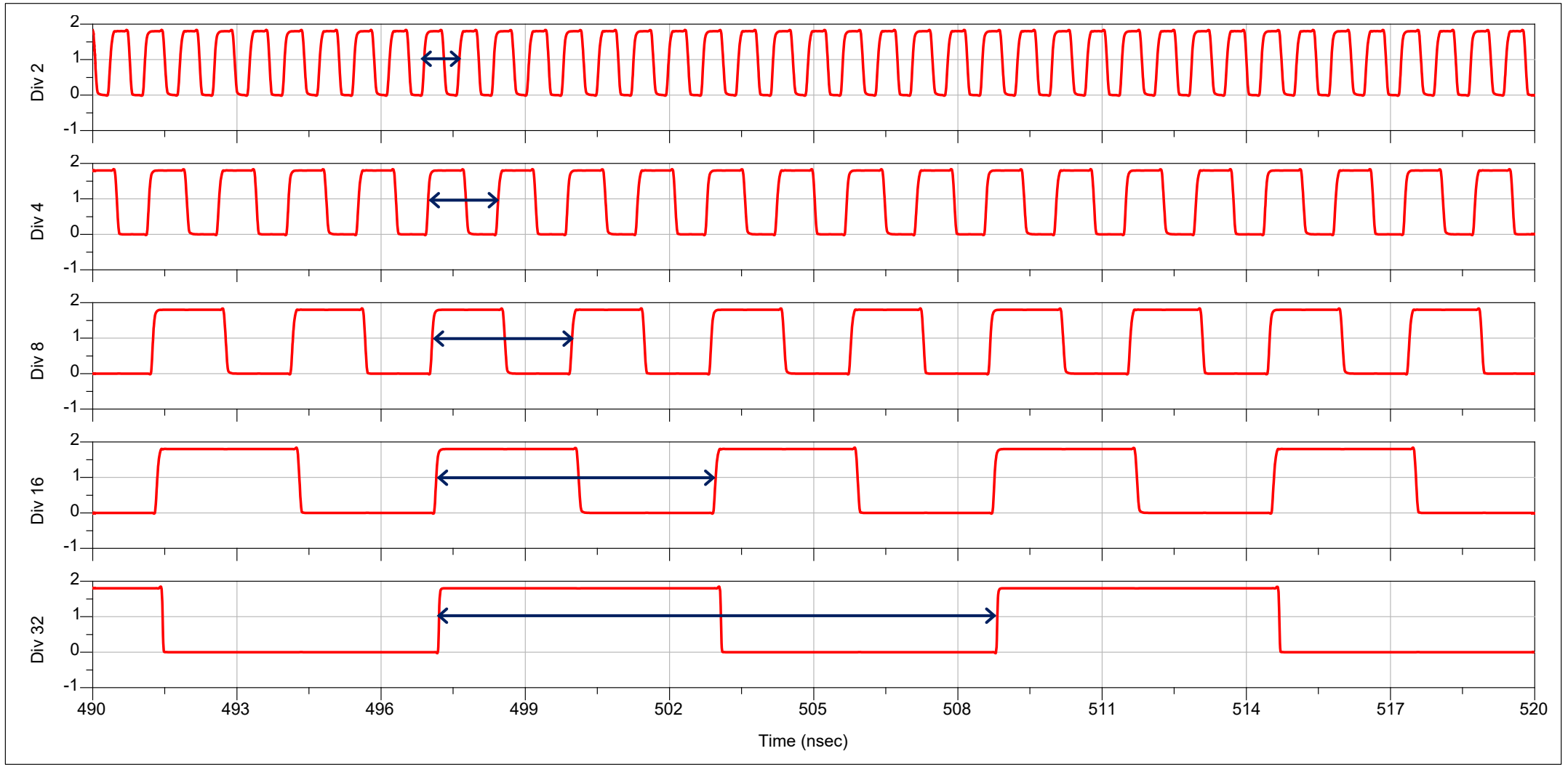


Fig 36. TSPC divider chain output signal

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- **Varactor Delay Element**
- Layout & Measurement
- Conclusion & Reference

Structure of Varactor Delay Element

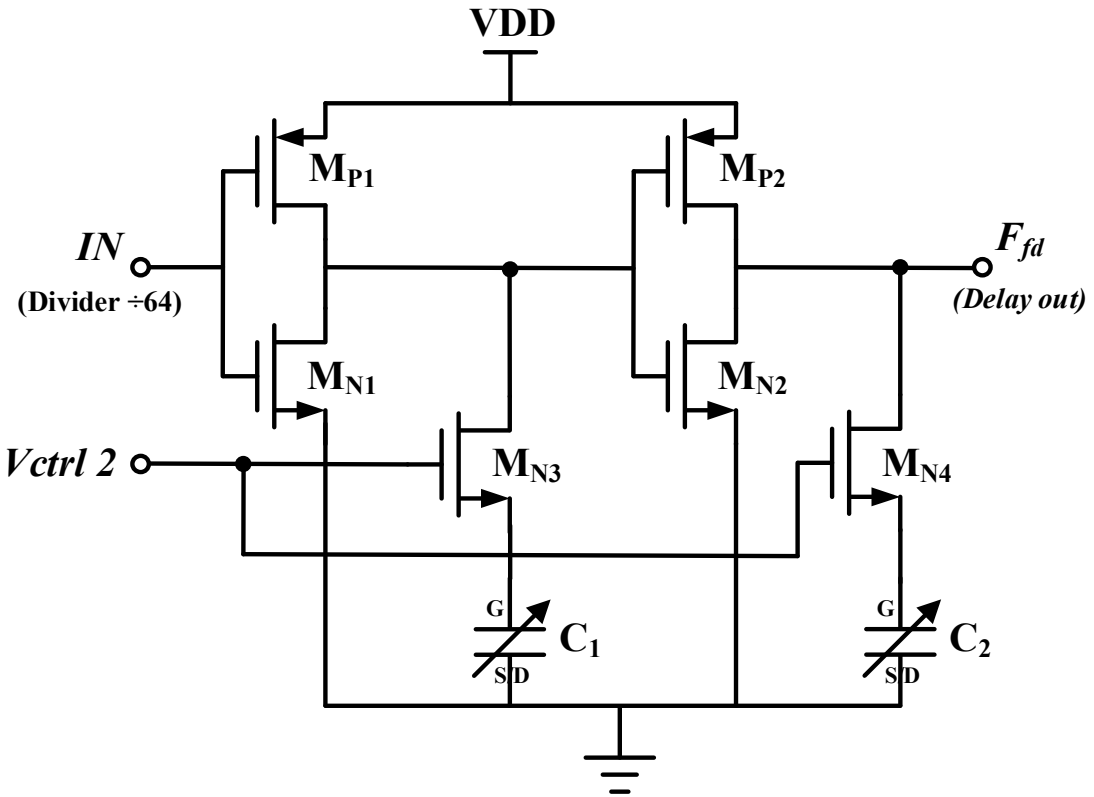


Fig 37. Structure of varactor delay element

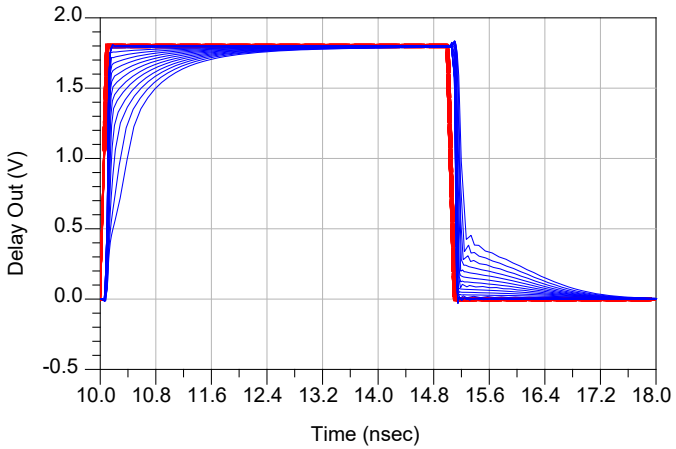


Fig 38. Delay curve of varactor

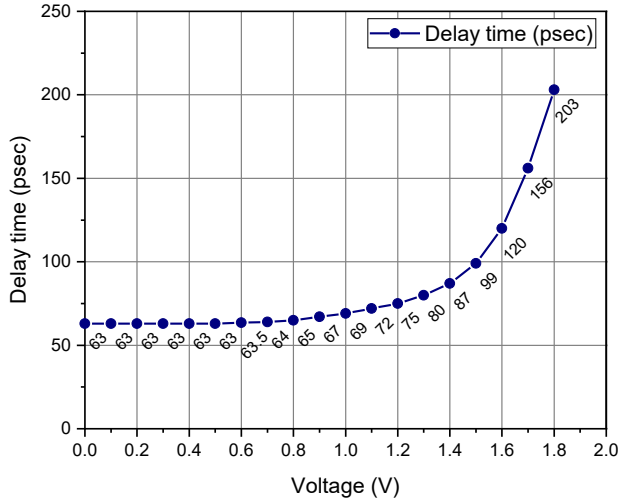


Fig 39. Delay time with control voltage

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- **Layout & Measurement**
- Conclusion & Reference

I/O Pin of Phase-Locked Loop System

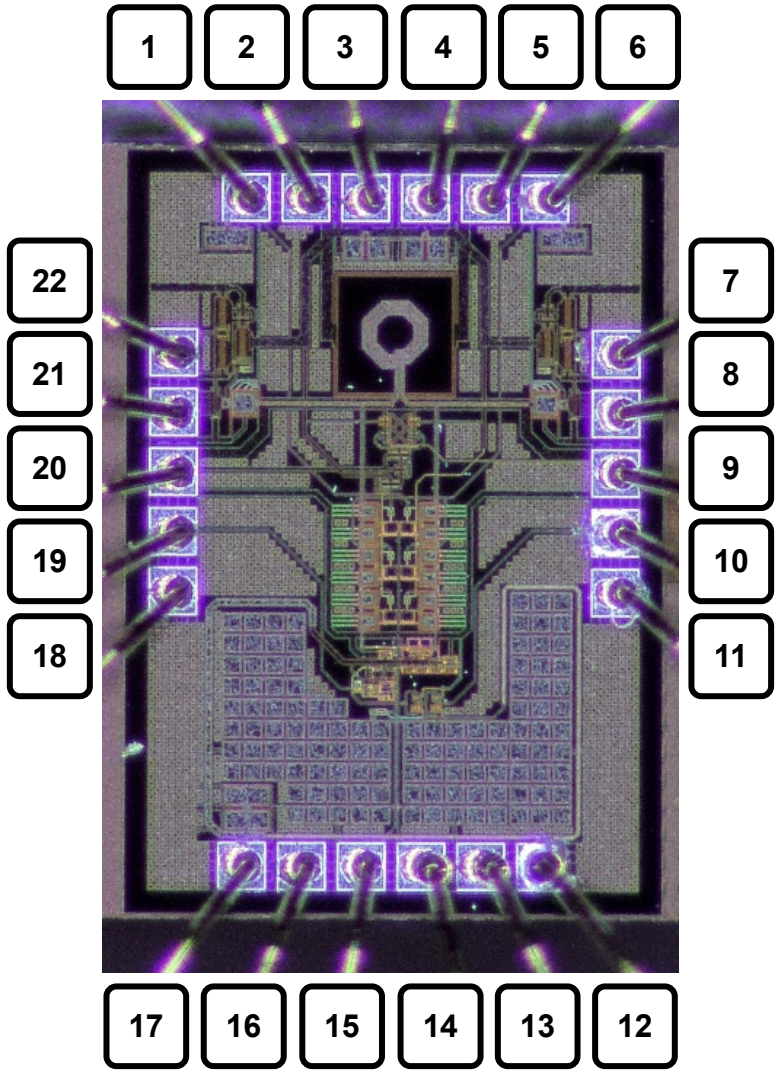


Fig 40. I/O Pin

Tab 2. I/O Pin of Phase-Locked Loop System

1	VB1	Supply for Array of Varactors	12	VDD	Supply for Digital Circuit
2	GND	Ground for chip	13	GND	Ground for chip
3	VDD	Supply for Analog Circuit	14	Vctrl	Test for Vctrl
4	VDD	Supply for Analog Circuit	15	Vctrl2	Supply for Delay Element
5	GND	Ground for chip	16	GND	Ground for chip
6	VB2	Supply for Array of Varactors	17	VDD2	Supply for Digital Circuit
7	GND	Ground for chip	18	GND	Ground for chip
8	D2	Output for VCO	19	Fref	Reference frequency
9	GND	Ground for chip	20	GND	Ground for chip
10	Ffd	Test for feed back	21	D1	Output for VCO
11	GND	Ground for chip	22	GND	Ground for chip

Post Simulation of Phase-Locked Loop System

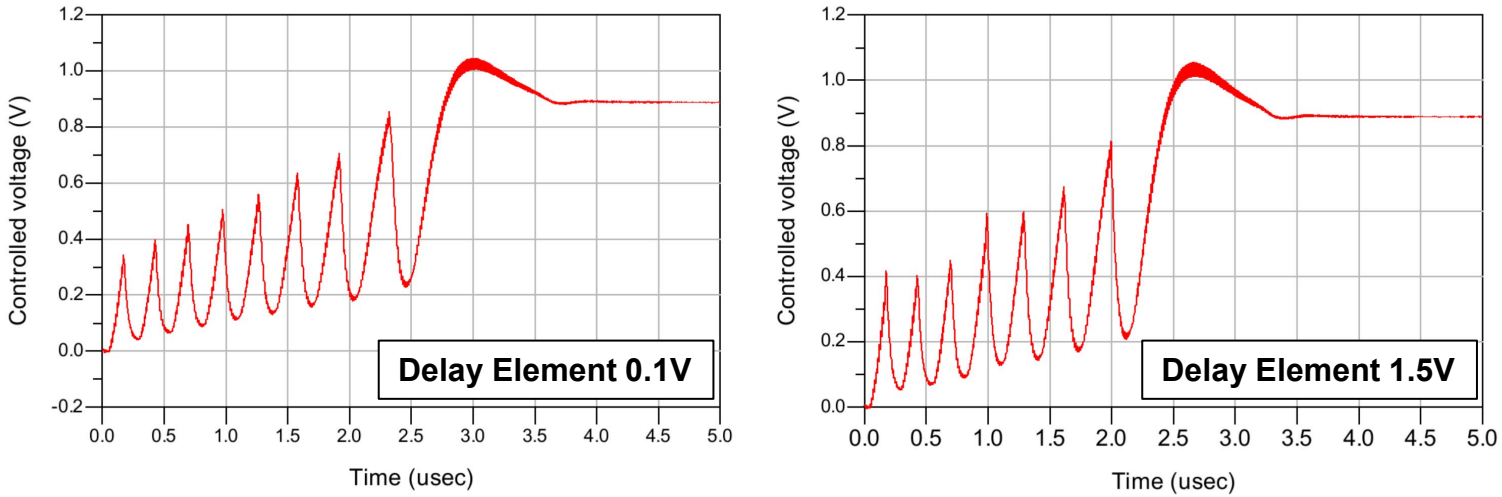


Fig 41. Phase-locked loop system control voltage

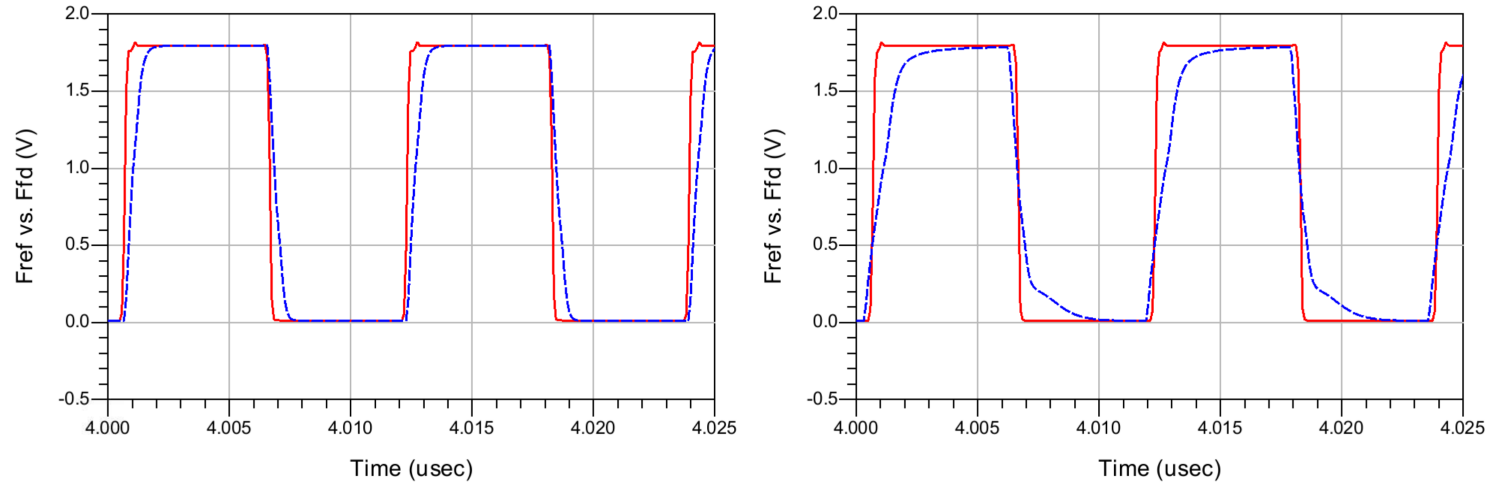


Fig 42. Feedback signal *Ffd*

Post Simulation of Phase-Locked Loop System

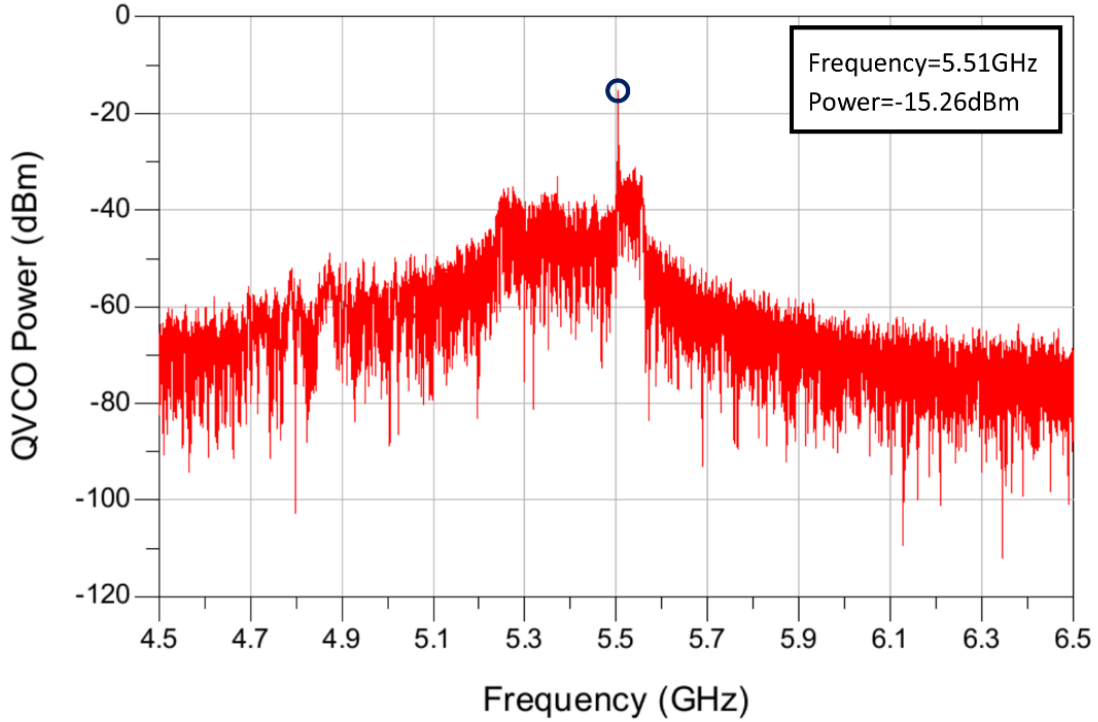


Fig 43. VCO circuit output power when PLL locked

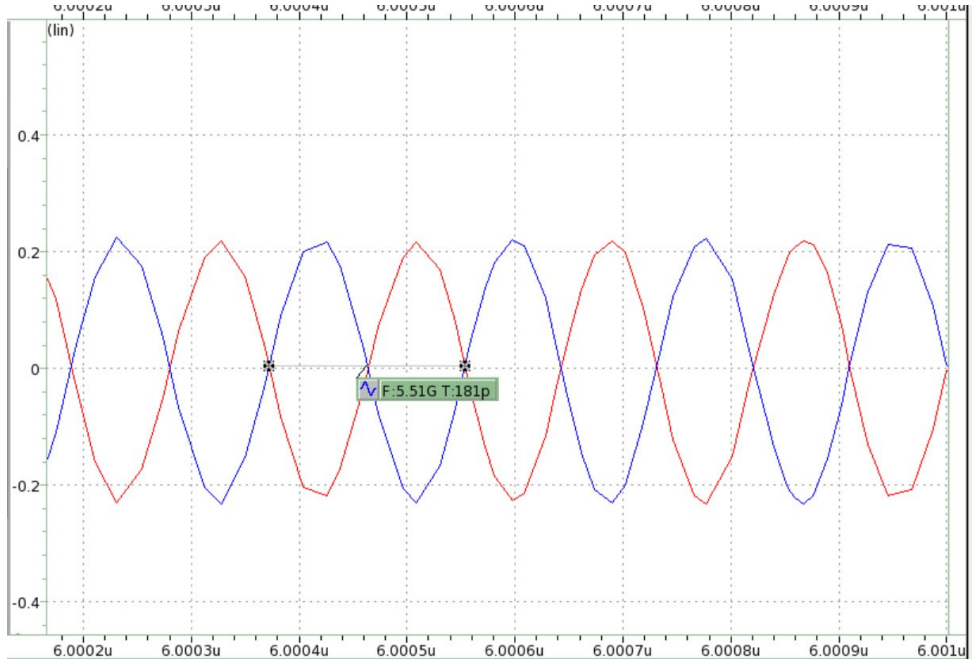


Fig 44. VCO circuit output signal when PLL locked

PCB Board with Two Phase-Locked Loop Chips

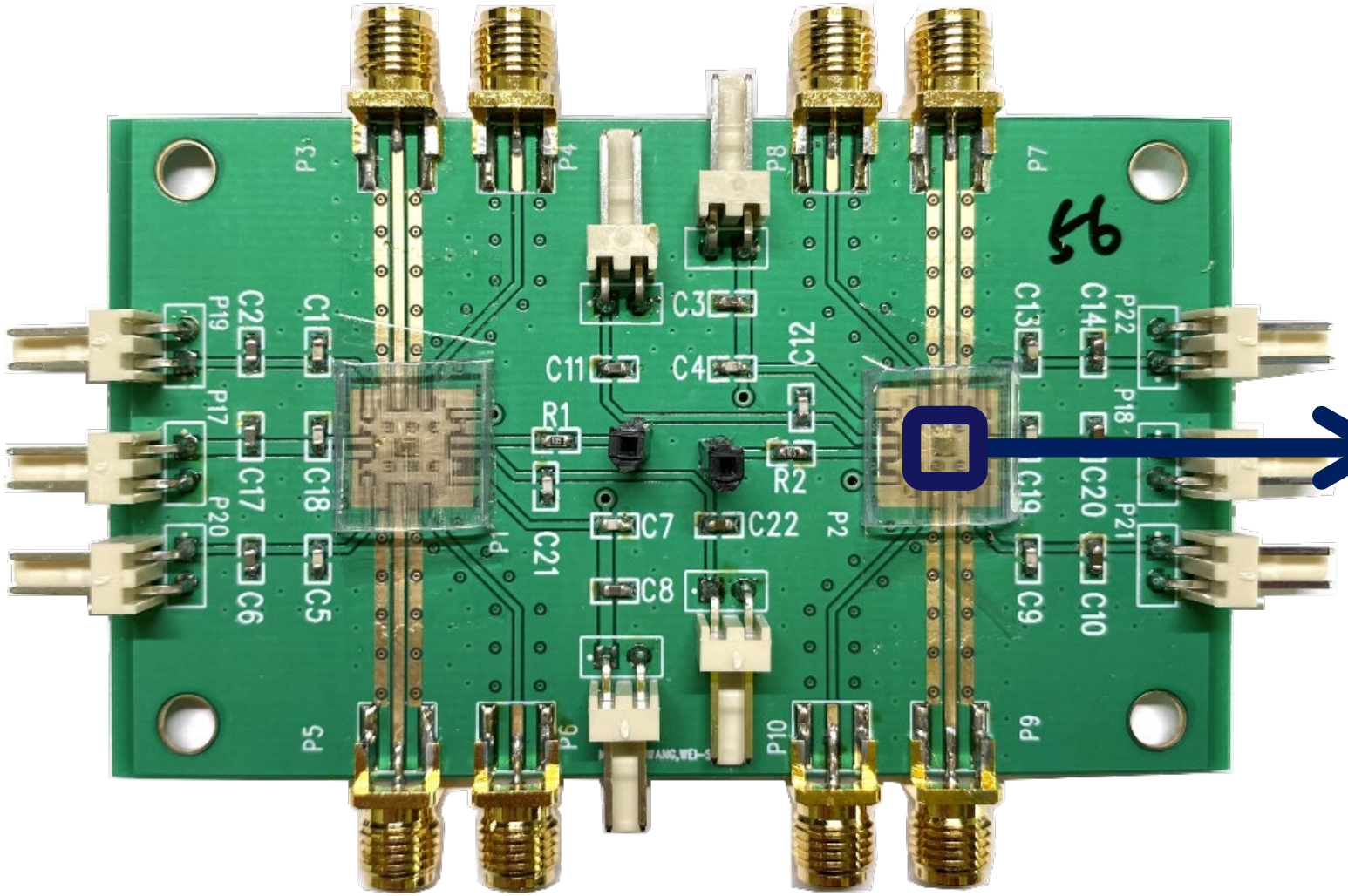


Fig 45. PCB board

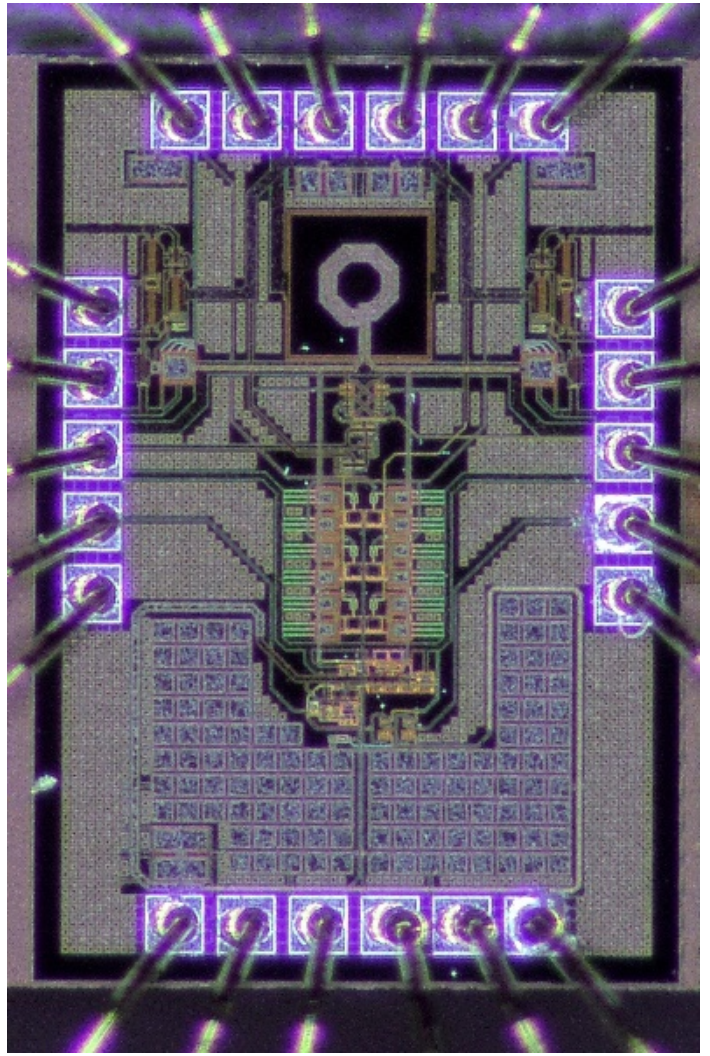


Fig 46. Photomicrograph of phase locked loop chip

PCB Board with Two Phase-Locked Loop Chips

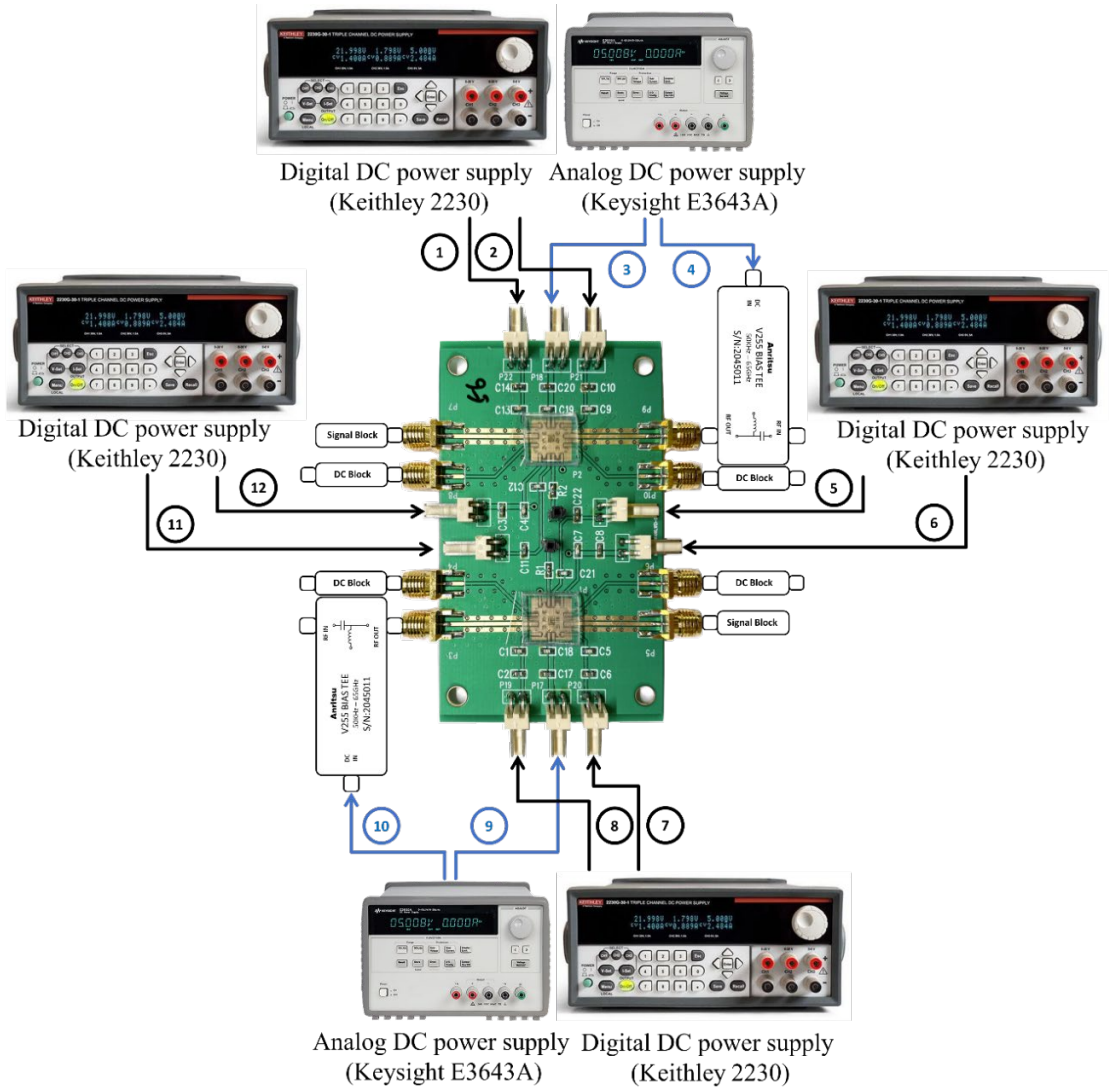


Fig 47. PCB Board power terminal of measurement

Tab 3. PCB Board analog power terminal of measurement

Number	3	4	9	10
Analog (V)	1.8	0.9	1.8	0.9

Tab 4. PCB Board digital power terminal of measurement

Number	1	2	5	6	7	8	11	12
Digital (V)	1.2	0.6	0-1.5	1.8	1.2	0.6	0-1.5	1.8

PCB Board with Two Phase-Locked Loop Chips

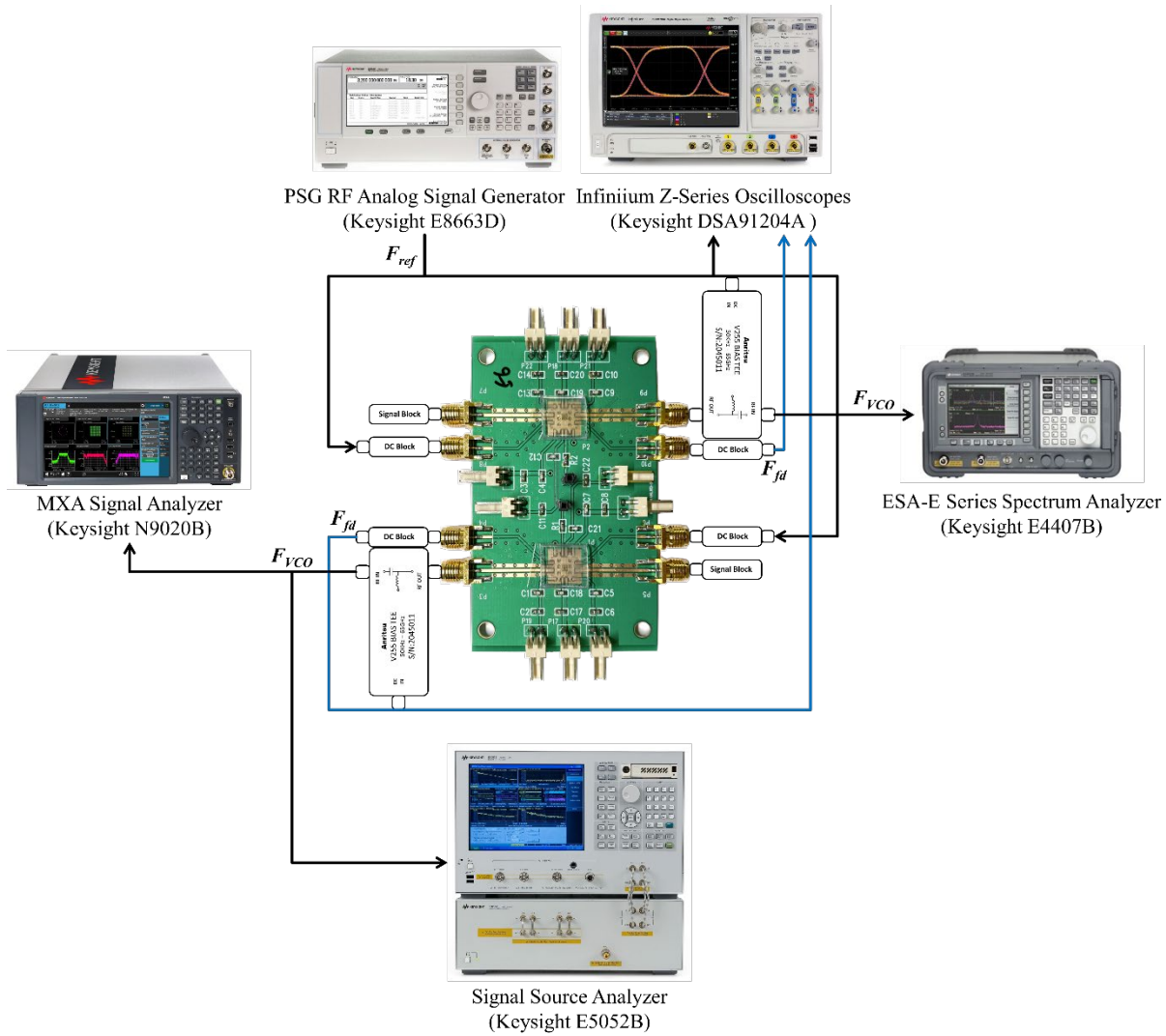


Fig 48. PCB Board signal terminal of measurement

Measurement

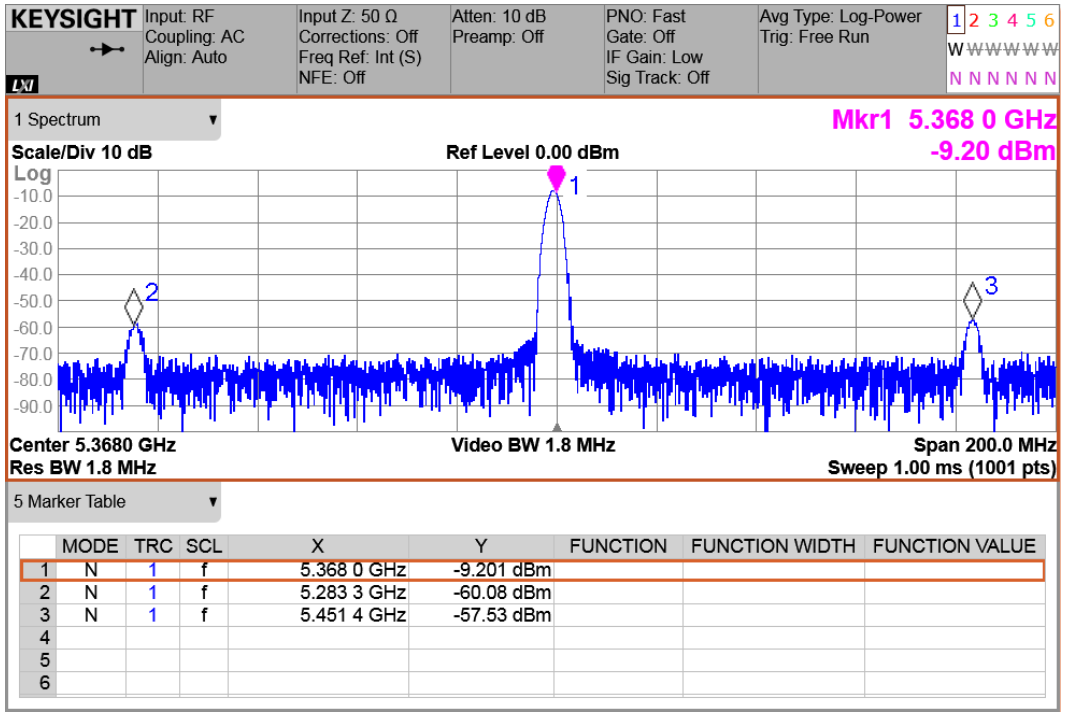


Fig 49. VCO1 circuit output power

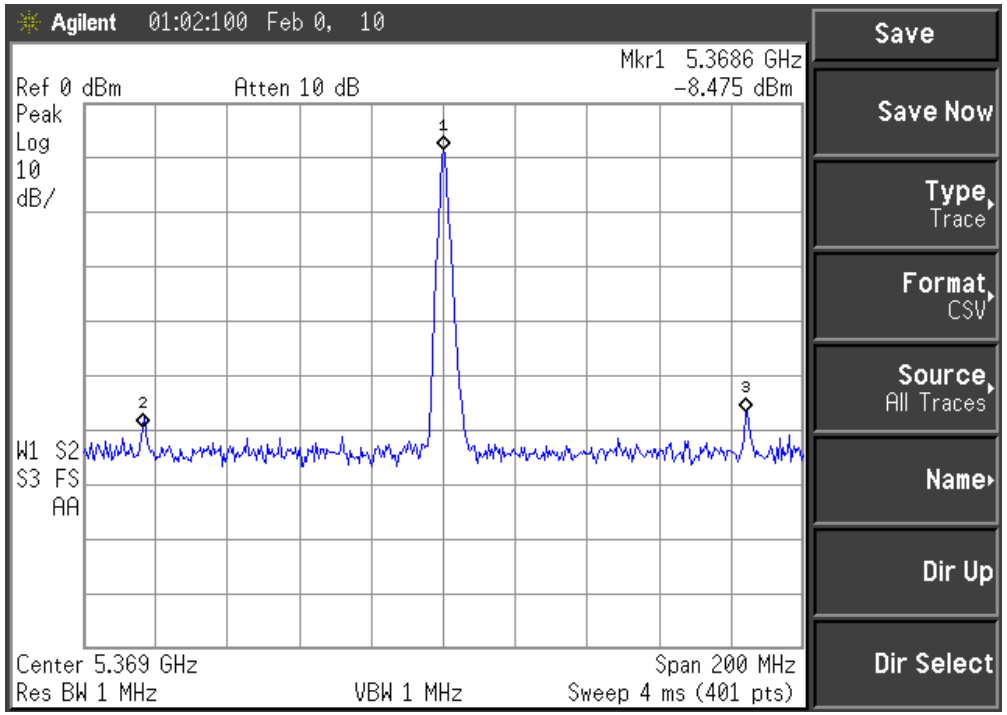


Fig 50. VCO2 circuit output power

Measurement

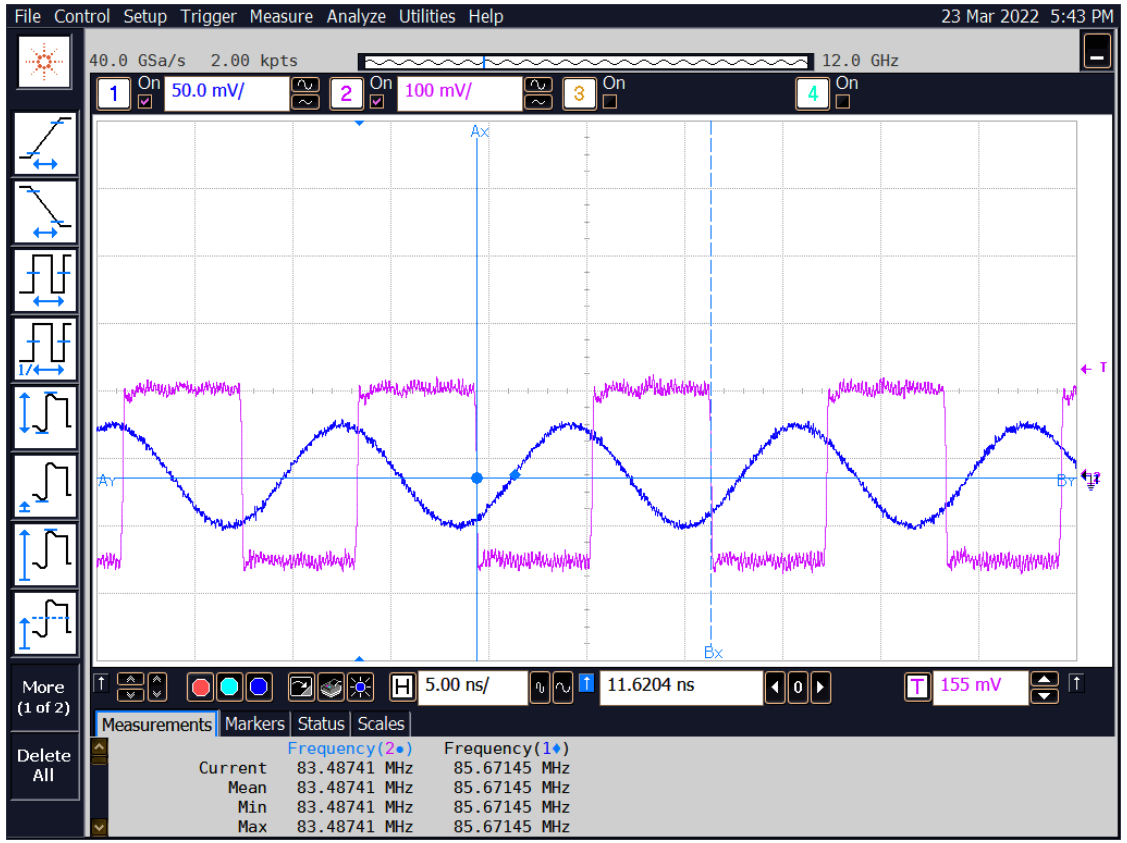


Fig 51. F_{ref} and F_{fd} of VCO circuit

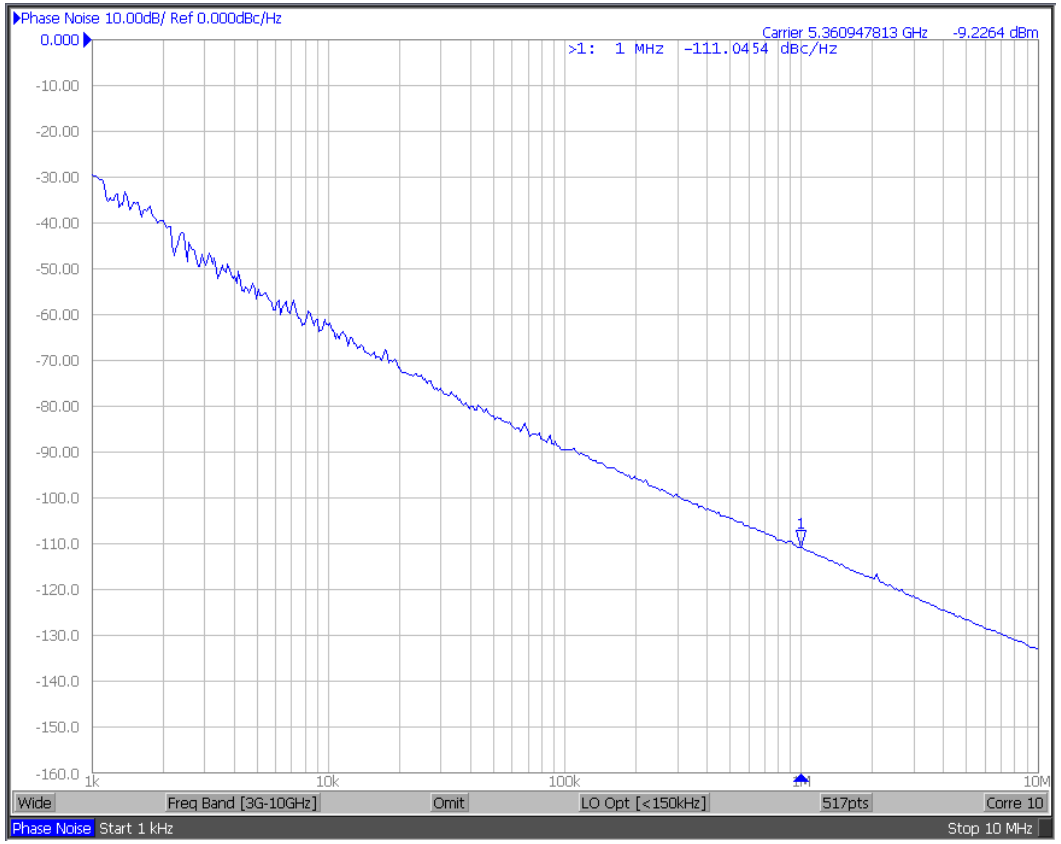


Fig 52. Phase noise of VCO circuit

Comparison Table

Tab 5. Circuit performance comparison Table

	This work	[35]	[36]	[37]
CMOS Tech. (nm)	180	180	180	130
Supply Voltage	1.8 V	1.8 V	1.8 V	1.2 V
Current Dissipation	9.5 mA	11 mA	9.5 mA	22.7 mA
Division	64	8/10	4	Frac-N
Reference Frequency (MHz)	83.75-92.96	10	4.7	50
Operation Frequency (GHz)	5.36-5.95	5.27-5.6	4.9-5.3	2.3-2.4
Locking Time (ms)	< 6	< 20	<10	
Controllable Delay Time (ps)	63-203	-	-	-
Controlled-voltage of VCDL (V)	0~1.8	-	-	-
Phase Noise @1MHz (dBc/Hz)	-111	-114	-118	-120
Reference Spurs (dBc)	<-48	<-70	<-71	<-72
Chip Area (mm^2)	0.984	1.610	1.61	-

Layout of Quadrature Voltage Controlled Oscillator

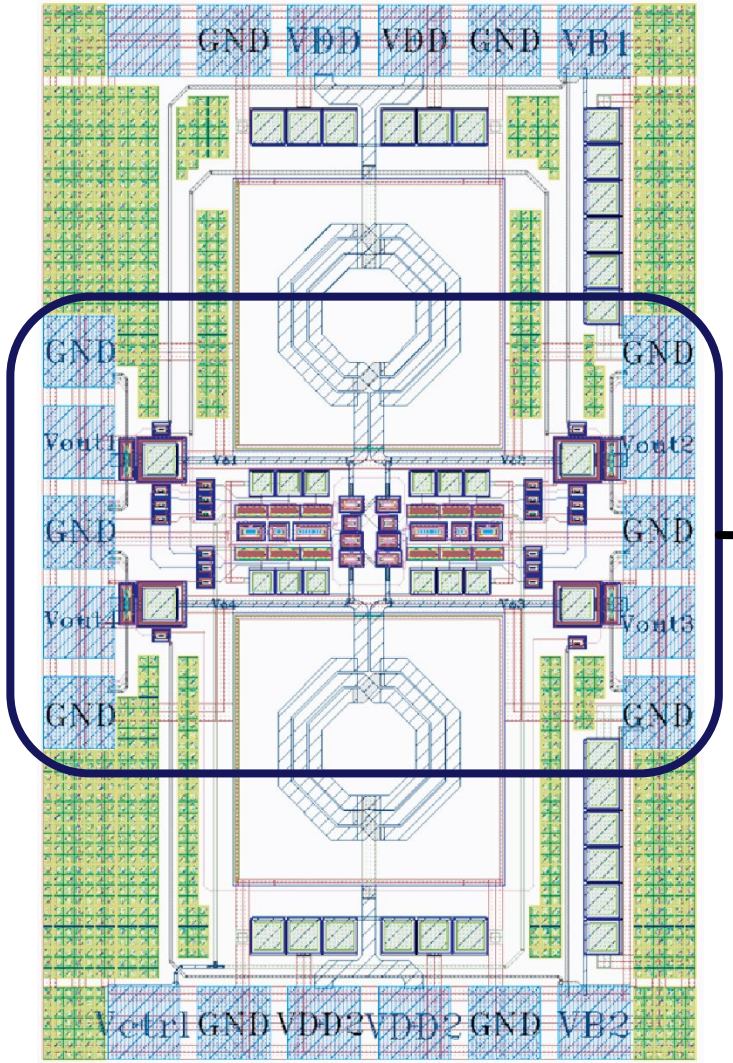


Fig 53. Layout of QVCO

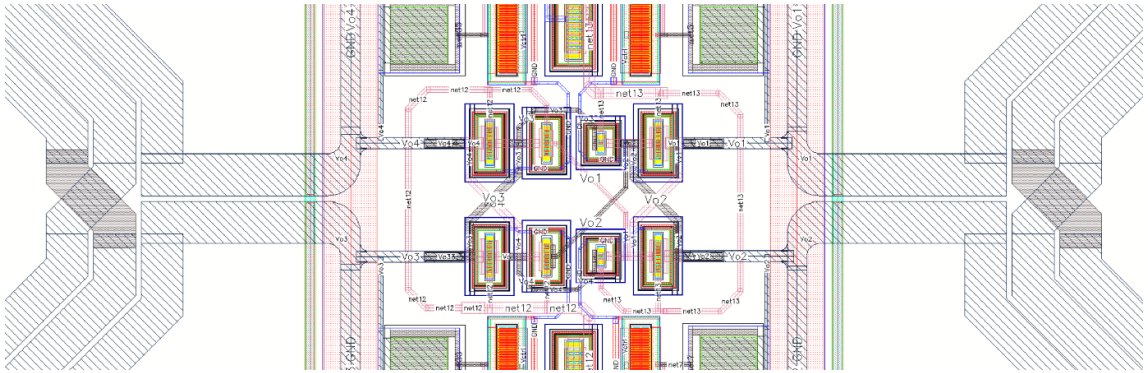


Fig 54. QVCO circuit in detail view

Post Simulation of Quadrature Voltage Controlled Oscillator

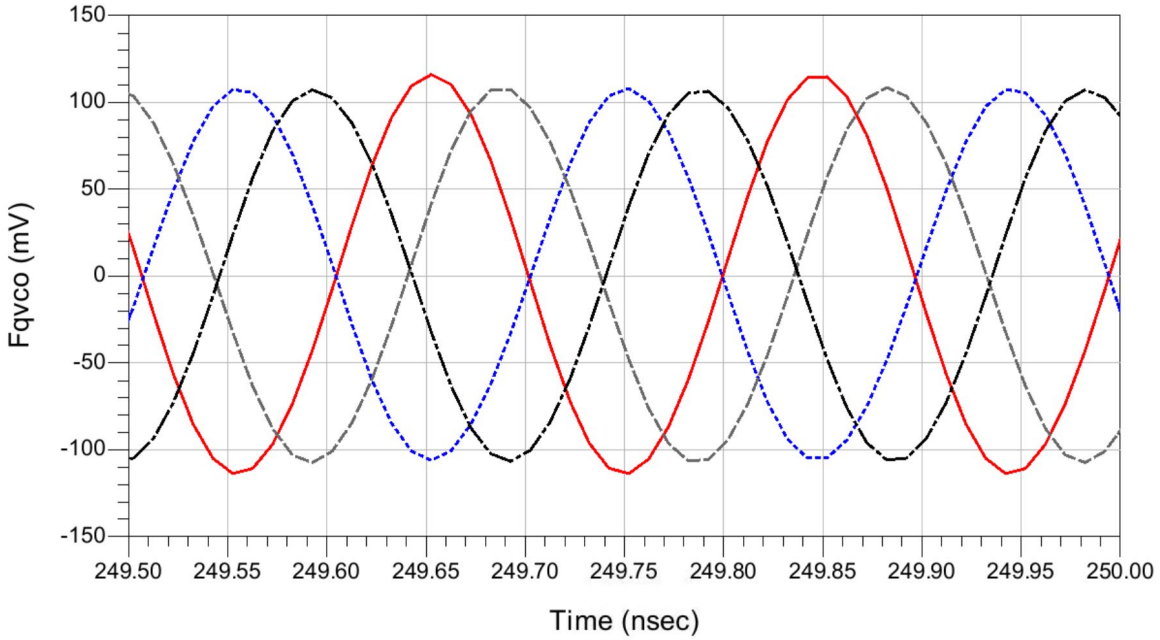


Fig 55. QVCO circuit output

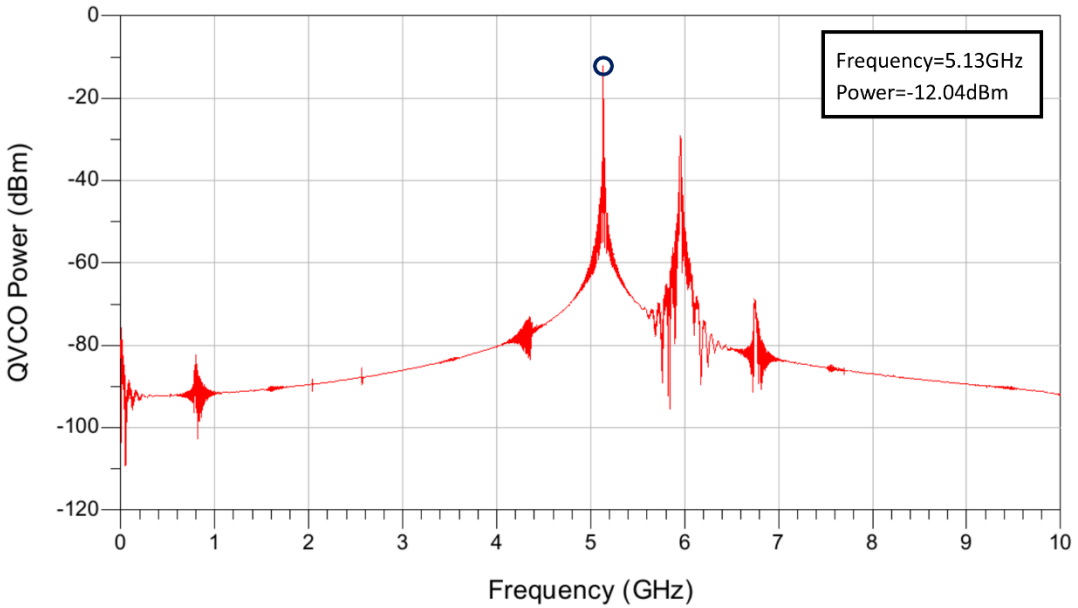


Fig 56. QVCO circuit output power

Post Simulation of Quadrature Voltage Controlled Oscillator

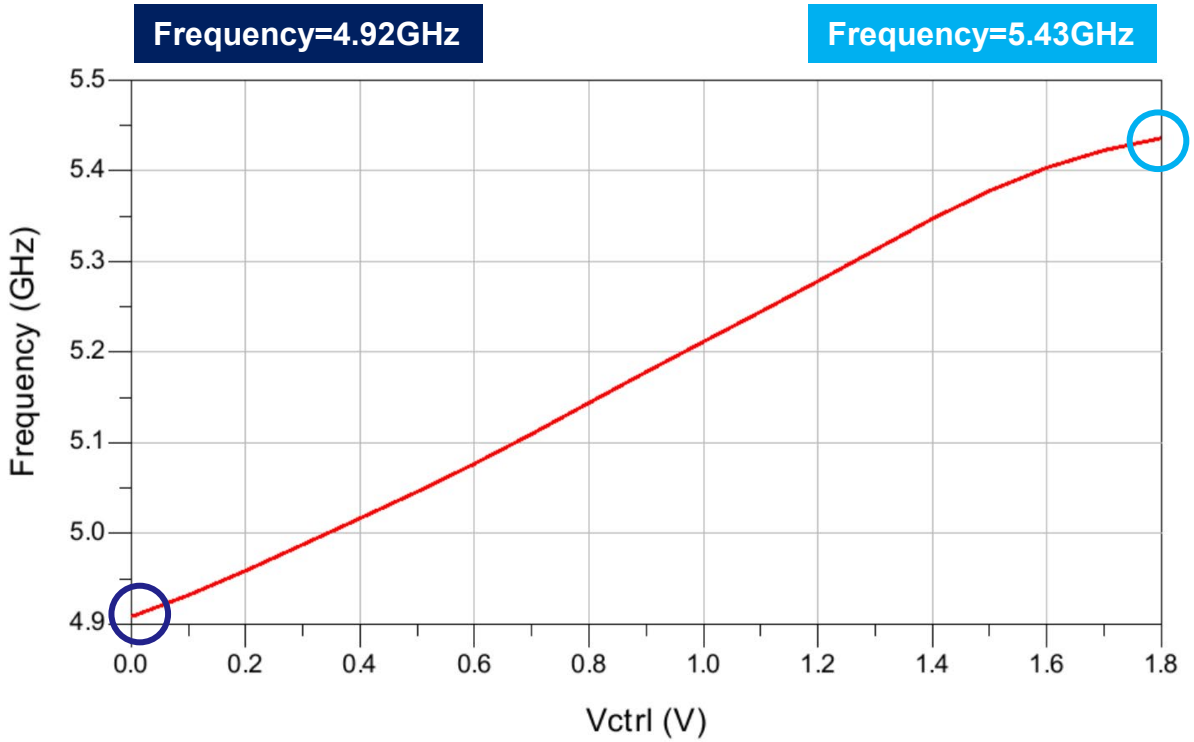


Fig 57. Tuning range of QVCO circuit

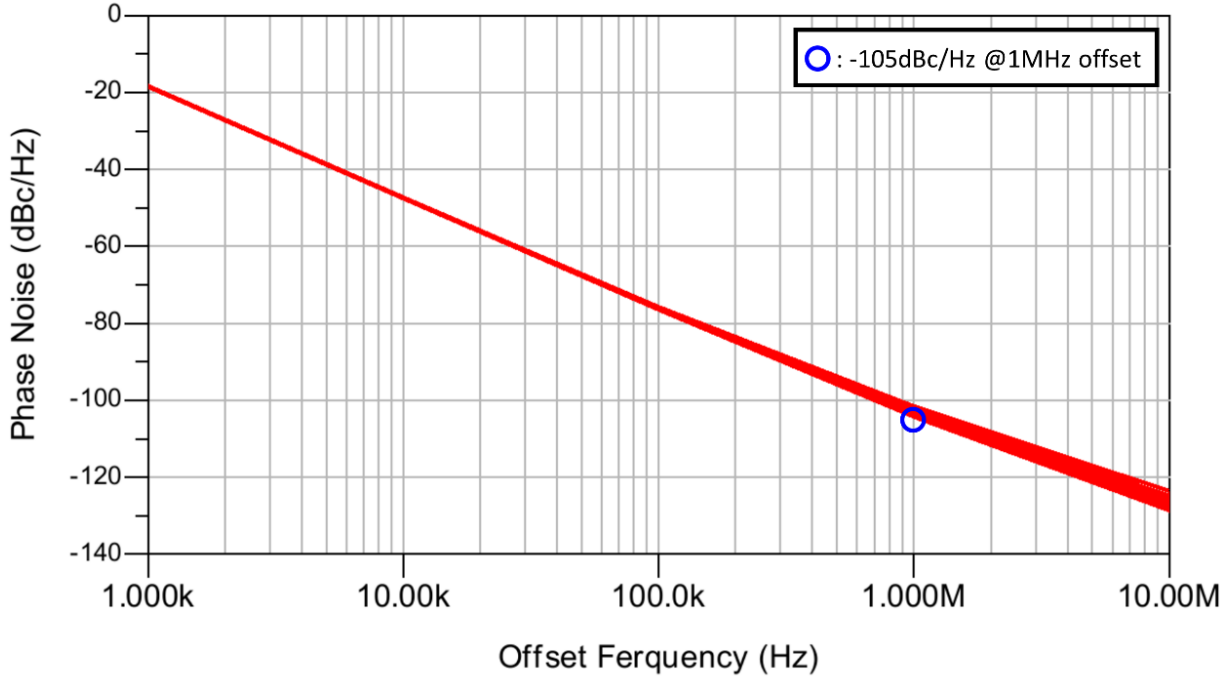


Fig 58. QVCO circuit phase noise

I/O Pin of Quadrature Voltage Controlled Oscillator

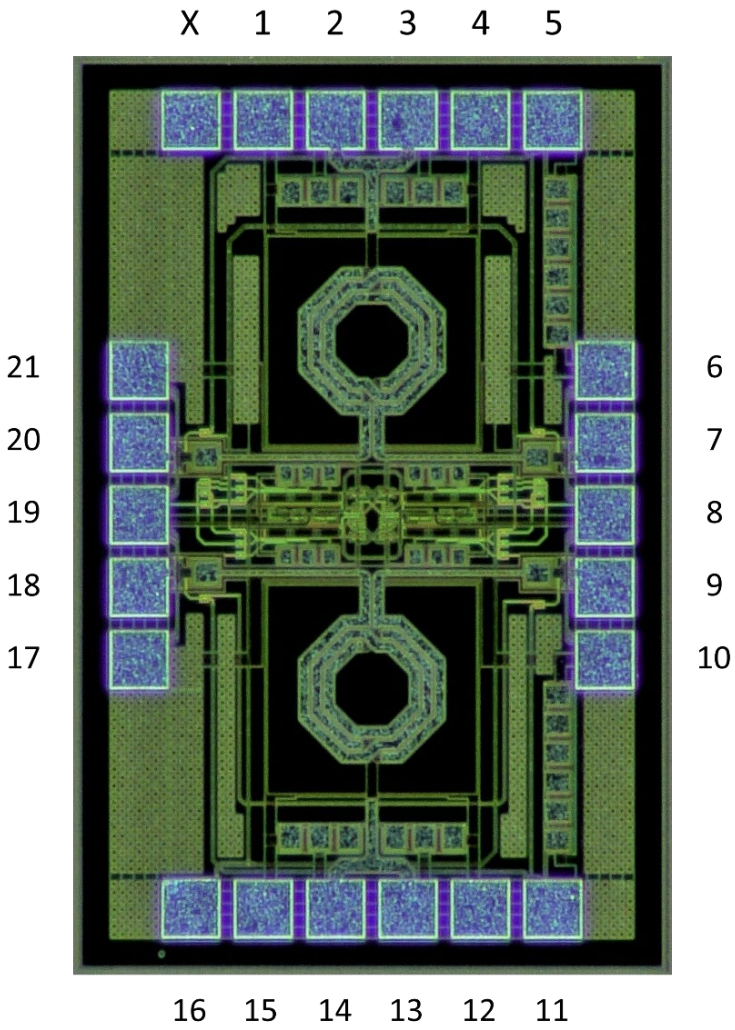


Fig 59. Photomicrograph of QVCO

Tab 6. 3GPP 5G NR FR1 add to Release 16

1	*GND*	Ground for chip	12	*GND*	Ground for chip
2	*VDD*	Supply for QVCO Circuit	13	*VDD2*	Supply for QVCO Circuit
3	*VDD*	Supply for QVCO Circuit	14	*VDD2*	Supply for QVCO Circuit
4	*GND*	Ground for chip	15	*GND*	Ground for chip
5	*VB1*	Bias for buffer	16	*Vctrl*	Test for Vctrl
6	*GND*	Ground for chip	17	*GND*	Ground for chip
7	*Vout2*	Output for QVCO	18	*Vout4*	Output for QVCO
8	*GND*	Ground for chip	19	*GND*	Ground for chip
9	*Vout3*	Output for QVCO	20	*Vout1*	Output for QVCO
10	*GND*	Ground for chip	21	*GND*	Ground for chip
11	*VB2*	Bias for buffer	X	*NO*	Unused

On Wafer Measurement of QVCO

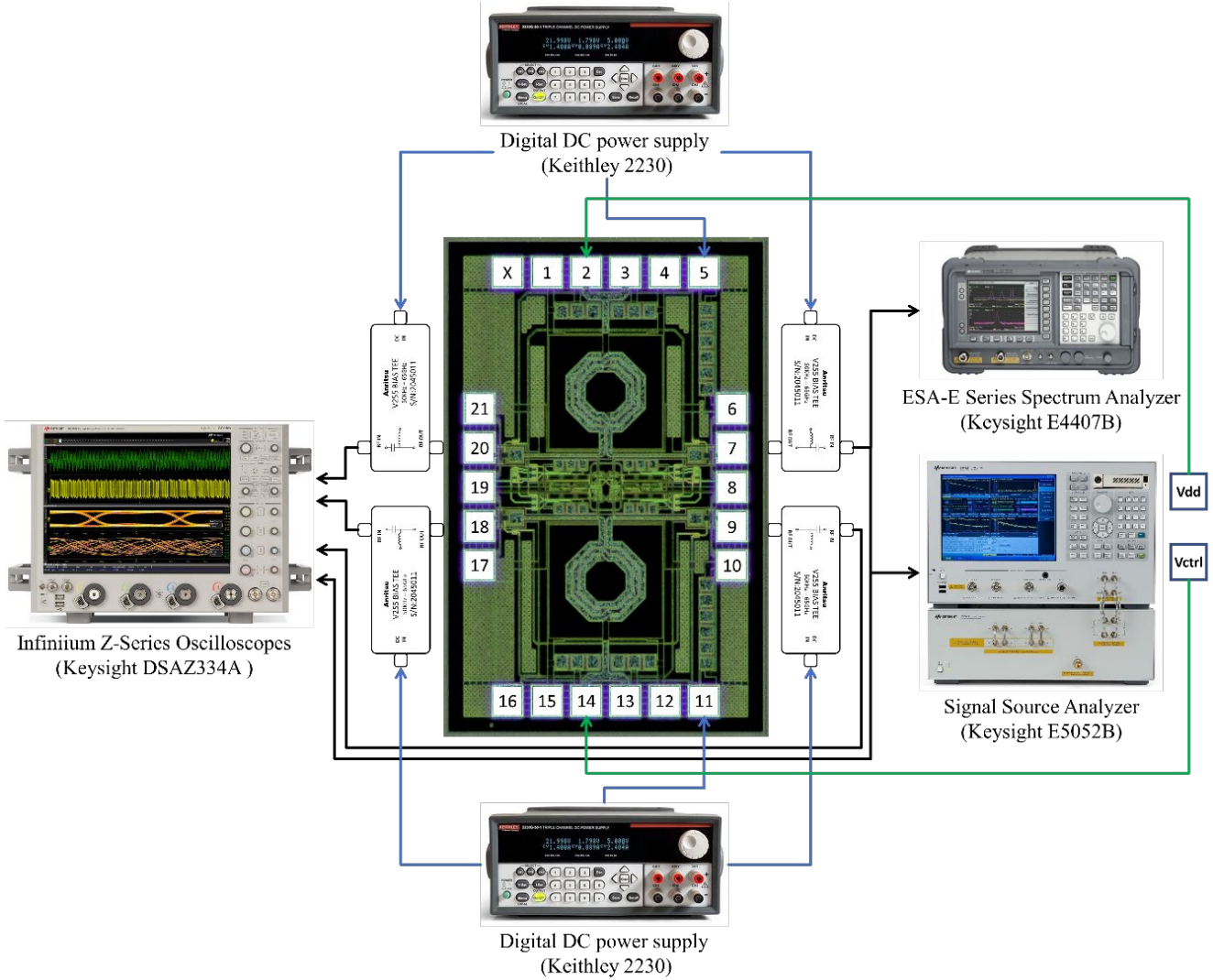


Fig 60. QVCO input power and output signal terminal of measurement

On Wafer Measurement of QVCO

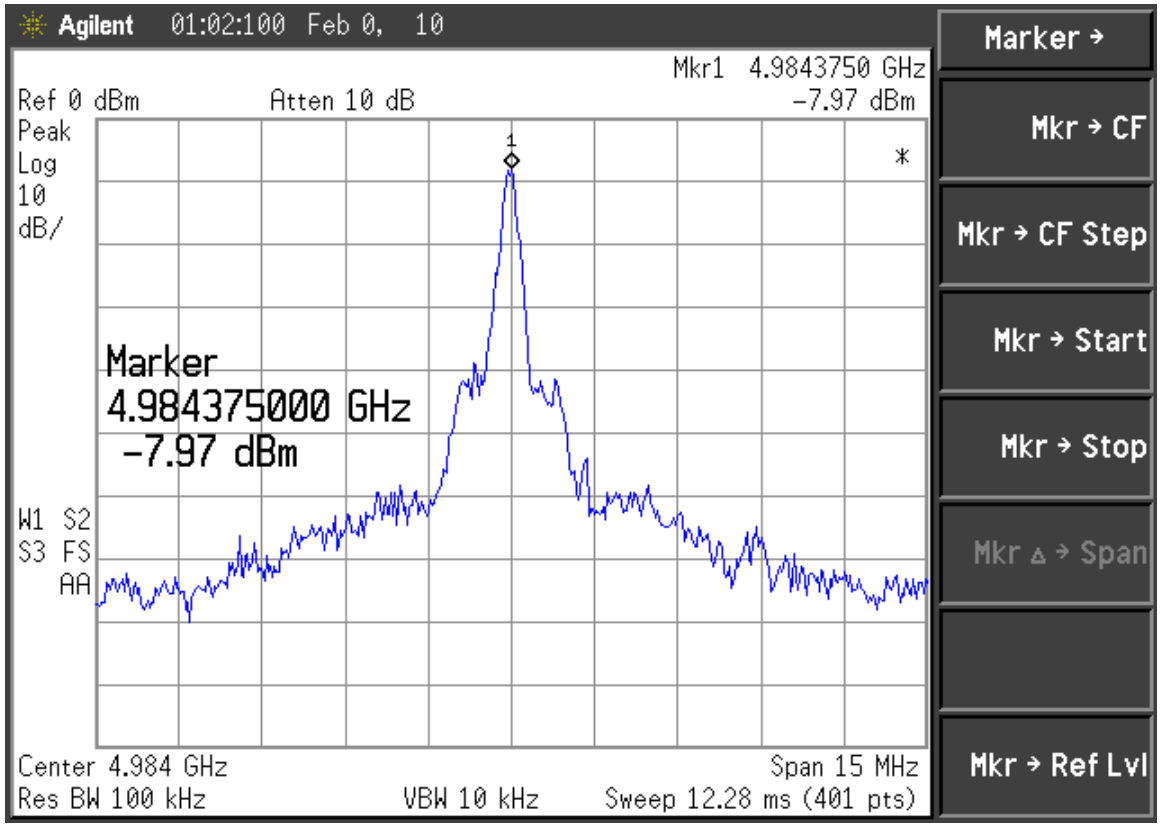


Fig 61. QVCO circuit output power

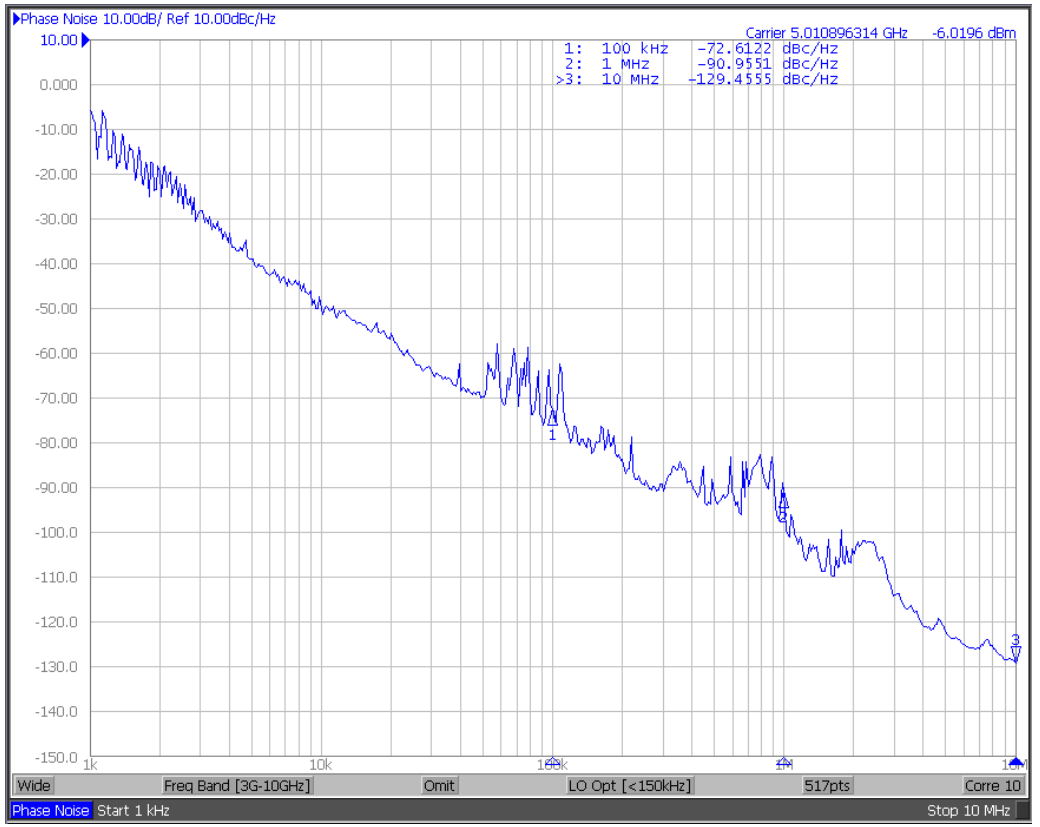


Fig 62. QVCO phase noise

On Wafer Measurement of QVCO

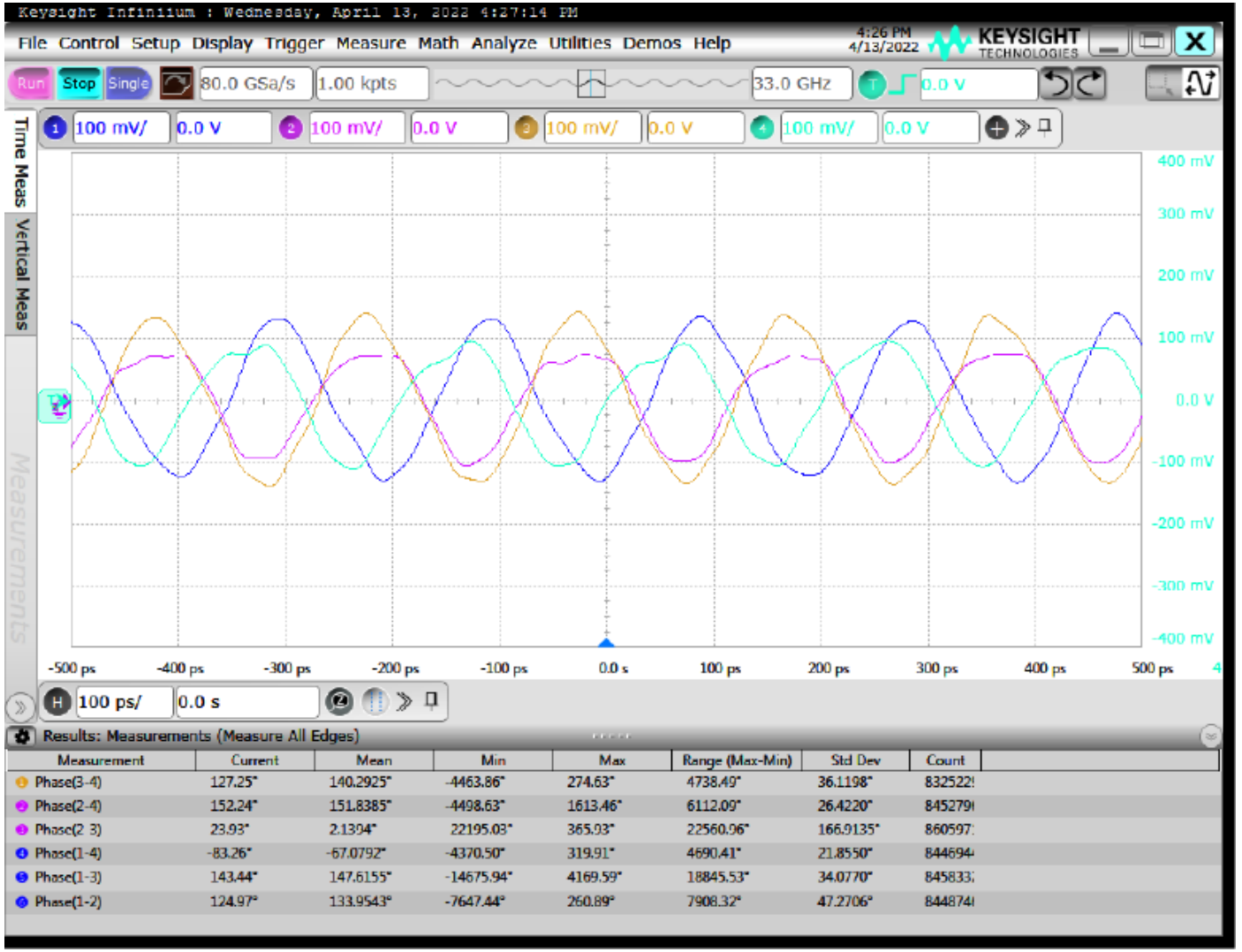


Fig 63. QVCO circuit output signal

Contents

- Introduction
- Related Technology
- Phase Frequency Detector & Charge pump
- Loop Filter
- Voltage Controlled Oscillator
- Quadrature-Voltage Controlled Oscillator
- Frequency Divider
- Varactor Delay Element
- Layout & Measurement
- **Conclusion & Reference**

Conclusion

- Promote the conformability of the chip
  - Put the loop filter in the chip promote the device integration.
  - Supply other control variable V_{ctrl2} to the communication system.

Reference

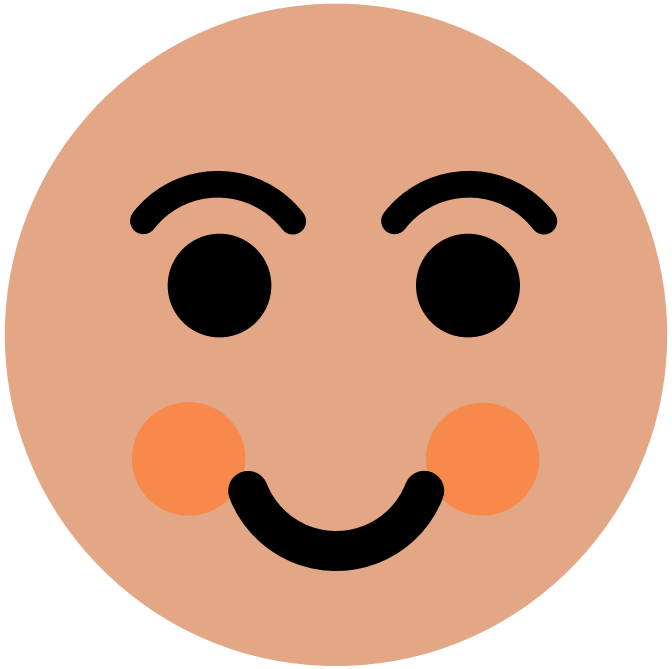
1. 3rd Generation Partnership Project, 3GPP Release 18. https://www.3gpp.org/release-18, Retrieved June 11, 2022.
2. Erik Dahlman, Stefan Parkvall and Johan Sköld, “5G NR The Next Generation Wireless Access Technology,” Academic Press, Elsevier Publication, ISBN:978-0-12-814323-0,2018.
3. Armstrong, Electrical World, 12 Dec. 1914 T. H. Lee, “The Design of CMOS Radio-Frequency Integrated Circuit,” Chap. 1, pp 15, Cambridge, 1998.
4. 張傳福, 趙立英, 張宇, “最完整5G技術架構白皮書”, 深智數位股份有限公司,ISBN:978-986-97726-9-3, 2020.
5. 張軍民, 金超, 蔣伯章, “新一代5G行動網路最佳化進階實戰”, 深智數位股份有限公司, ISBN:978-986-0776-43-0, 2021.
6. A. Kaye and D. George, “*Transmission of Multiplexed PAM Signals Over Multiple Channel and Diversity Systems*,” IEEE Transactions on Communication Technology, vol.18, no. 5, pp. 520-526, October 1970.
7. L. H. Brandenburg and A. D. Wyner, “Capacity of the Gaussian channel with memory: The multivariate case”, The Bell System Technical Journal, vol.53, no.5, pp.745-778, May-June, 1974.
8. W. van Etten, “*Maximum Likelihood Receiver for Multiple Channel Transmission Systems*,” IEEE Transactions on Communications, vol. 24, no. 2, pp. 276-283, February, 1976.
9. J.Salz, “Digital transmission over cross-coupled linear channels,” AT&T Technical Journal, vol. 64, no. 6, pp. 1147-1159, July-Aug. 1985.
10. Paulraj Arogyaswami J and Kailath Thomas, “Increasing capacity in wireless broadcast systems using distributed transmission directional reception (DTDR),” US patent: US5345599A, 1994.
11. C. E. Shannon, “A mathematical theory of communication,” The Bell System Technical Journal, vol. 27, no. 3, pp. 379-423, July 1948.
12. L L Peterson and B S Davie, Computer Networks: a systems approach (Morgan Kaufmann), pp 94-95, ISBN: 1-55860-368-9(Paperback ISBN: 1-558 60-404-9), 1996.
13. International Telecommunication Union (ITU), ITU Radio Regulations, Volume 1, Article 2; Edition of 2020. Available online at “Article 2.1: Frequency and wave-Length bands”. Radio Regulations 2016 Edition. January 1 2017. Retrieved June 9, 2022.

Reference

14. IEEE, "*IEEE Standard Letter Designations for Radar-Frequency Bands*," in IEEE Std 521-2002 (Revision of IEEE Std 521-1984) , vol., no., pp.1-10, Jan 8 2003.
15. 3rd Generation Partnership Project, 3GPP Release 15. https://www.3gpp.org/release-15 Retrieved June 19, 2022.
16. 3rd Generation Partnership Project, 3GPP Release 16. https://www.3gpp.org/release-16 Retrieved June 19, 2022.
17. Qualcomm Inc., Snapdragon X60 5G Modem-RF System, https://www.qualcomm.com/products/technology/modems/snapdragon-x60-5g-modem, Retrieved June 20, 2022.
18. Apple Inc., 5G and LTE. Find the iPhone that's right for your country or region., https://www.apple.com/iphone/cellular/, Retrieved June 20, 2022.
19. ITU-R, IMT Vision—Framework and overall objectives of the future development of IMT for 2020 and beyond. Recommendation ITU-R M.2083, September 2015.
20. Alan Douglas, Who Invented the Superheterodyne? Originally published in the proceedings of the Radio Club of America, Nov. 1990, Vol.64 no.3 "The Legacies of Edwin Howard Armstrong.," https://antiqueradios.com/superhet/, Retrieved June 28, 2022.
21. H. De Belleszice, La reception synchrone, L'onde e'lectrique, vol. 11, pp.225-240,1932.
22. Behzad Razavi, RF Microelectronics (2nd Edition), Prentice Hall Press, ISBN: 978-0137134731, 2011.
23. 高曜煌, "射頻鎖相迴路IC設計(第二版)", 滄海圖書資訊股份有限公司, ISBN:978-986-5647-76-6, 2017.
24. 遠坂俊昭, 宋國明, "PLL電路設計及應用" , 全華科技圖書股份有限公司, ISBN:978-957-21-5526-4, 2006.
25. 劉深淵, 楊清淵, "鎖相迴路" , 滄海圖書資訊股份有限公司, ISBN:978-986-7287-91-5, 2020.
26. Won-Hyo Lee, Jun-Dong Cho and Sung-Dae Lee, "A high speed and low power phase-frequency detector and charge-pump," Proceedings of the ASP-DAC '99 Asia and South Pacific Design Automation Conference 1999 (Cat. No.99EX198), pp.269-272 vol.1, 1999.
27. W. Rhee, "*Design of high-performance CMOS charge pumps in phase-locked loops*," 1999 IEEE International Symposium on Circuits and Systems (ISCAS), pp. 545-548 vol.2, 1999.

Reference

28. 張簡協修, 運用0.18 um CMOS 製程研製2.4 GHz可調式雙模態主動濾波器及全積體化頻率合成器, 國立交通大學電信工程研究所碩士論文, 民國101年。
29. 邱顯鴻, 低電壓互補式金氧半導體高頻壓控振盪器與鎖相迴路設計, 國立交通大學電機學院電信學程碩士論文, 民國96年。
30. 黃子哲, 尾電流塑形低相位雜訊暨電流共用低功率四相位輸出壓控振盪器設計與研究, 國立交通大學電信工程研究所碩士論文, 民國98年。
31. 張心瑜, 具自動頻率校正之5GHz頻率合成器, 國立臺北大學電機工程學系研究所碩士論文, 民國109年。
32. D. B. Leeson, "*A simple model of feedback oscillator noise spectrum*," in Proceedings of the IEEE, vol. 54, no. 2, pp. 329-330, Feb. 1966.
33. A. Hajimiri and T. H. Lee, "*A general theory of phase noise in electrical oscillators*," in IEEE Journal of Solid-State Circuits, vol. 33, no. 2, pp. 179-194, Feb. 1998.
34. P. Raha, S. Randall, R. Jennings, B. Helmick, A. Amerasekera and B. Haroun, "*A robust digital delay line architecture in a 0.13 /spl mu/m CMOS technology node for reduced design and process sensitivities*," Proceedings International Symposium on Quality Electronic Design, pp. 148-153 , 2002.
35. W. -H. Chiu, Y. -H. Huang and T. -H. Lin, "*A Dynamic Phase Error Compensation Technique for Fast-Locking Phase-Locked Loops*," in IEEE Journal of Solid-State Circuits, vol. 45, no. 6, pp. 1137-1149, June 2010.
36. W. Chiu, Y. Huang and T. Lin, "*A 5GHz phase-locked loop using dynamic phase error compensation technique for fast settling in 0.18-μm CMOS*," 2009 Symposium on VLSI Circuits, pp. 128-129, 2009.
37. D. Liao, F. F. Dai, B. Nauta and E. Klumperink, "*Multi-phase sub-sampling fractional-N PLL with soft loop switching for fast robust locking*," 2017 IEEE Custom Integrated Circuits Conference (CICC), pp. 1-4, 2017.



Thank you for your attention !!!